

Carbon-Aware Scheduling for Data Centres

*Reducing Scope 2 Carbon Emissions through Forecast-Driven
Temporal Workload Shifting*

A Complete Project Report

Module	IS6611 — Applied Research in Business Analytics
Institution	Cork University Business School, University College Cork
Group	11
Data Sources	ORNL Summit Supercomputer · Tennessee Valley Authority (EIA)
Deliverable	End-to-end analytics pipeline + live dashboard

Group Members: Aditya Anil More · Chetan Dummegere Kumar · Jobi Joy Mathew · Kartik Anil Shah · Sarvesh Deepak Pisal · Srushti Rajendrakumar Shetti

Table of Contents

1. Executive Summary	3
2. Introduction and Background	4
3. Problem Statement and Objectives	6
4. Literature Review	8
5. Solution Architecture and Methodology	10
6. Data Acquisition	11
7. Data Integration	14
8. Descriptive Analysis	17
9. Predictive Forecasting	22
10. Prescriptive Optimisation	27
11. Delivery and Deployment	30
12. Consolidated Findings and Insights	32
13. Business Value	33
14. Limitations	36
15. Future Work	37
16. Conclusion	38
17. References	39
18. Appendix A — Project Structure and Reproducibility	40
19. Appendix B — Decisions Log	41
20. Appendix C — Key Configuration and Results	42
21. Appendix D — Data Dictionaries	43
22. Appendix E — Worked Examples	45
23. Appendix F — Glossary	46

1. Executive Summary

Data centres and high-performance computing (HPC) facilities consume electricity around the clock. Globally they drew approximately 415 terawatt-hours in 2024 — about 1.5% of world electricity — and this is projected to double by 2030 as artificial-intelligence workloads accelerate. Yet the carbon footprint of that electricity is not constant: it changes every hour as the grid's generation mix shifts between clean sources (nuclear, hydro, wind, solar) and fossil fuels (coal, gas, oil). Despite this, the vast majority of facilities schedule their workloads on computational priority alone, with no awareness of grid conditions, so intensive jobs routinely run during the dirtiest hours of the day.

This project designs, builds, validates and deploys an end-to-end carbon-aware scheduling system. It forecasts grid carbon intensity up to 48 hours ahead and identifies the cleanest windows in which to run *delay-tolerant* workloads, reducing Scope 2 emissions at zero additional hardware cost and with no reduction in computation performed. The system is validated using real per-node power telemetry from the Oak Ridge National Laboratory (ORNL) Summit supercomputer paired with real generation data from the Tennessee Valley Authority (TVA) grid, retrieved from the U.S. Energy Information Administration (EIA) API.

The work spans the full analytics lifecycle: acquisition, integration, descriptive analysis, predictive forecasting, prescriptive optimisation and delivery through a live interactive dashboard. Along the way the project produced several findings of genuine analytical significance, three of which deserve emphasis:

- **The clean window moves unpredictably.** Averaged across four years, the spread between the cleanest and dirtiest hour of the day on the TVA grid is only about 8 gCO₂/kWh — almost flat. Yet *within any single day* the range averages 51 gCO₂/kWh. The single most frequently cleanest hour (09:00) is optimal on only 9.4% of days, so a fixed 'always run at hour X' rule misses the clean window on 90.6% of days. This proves that forecasting — not a static schedule — is required.
- **The best forecasting model is not the best scheduling model.** Of four models benchmarked, XGBoost achieved the lowest point-forecast error (MAE 17.95 gCO₂/kWh). But its recursive multi-step forecast collapses to a near-flat line, losing the daily *shape* that scheduling depends on — and when used to drive the scheduler it actually increased carbon. A simple shape-preserving forecast captured far more of the available saving. The metric that selects a forecaster (error) is not aligned with the objective that matters (ordering of clean hours).
- **Real constraints sharply reduce achievable savings — and honesty about this matters.** An unconstrained calculation suggested 314 tCO₂/year. Accounting for real cluster node capacity reduced the ceiling to 87 tCO₂/year, and accounting for forecast error reduced the realistic figure to roughly 34 tCO₂/year. Presenting this full hierarchy is more defensible than a single optimistic headline.

The final deliverable is a publicly deployed Streamlit dashboard with two stakeholder views — an Operations Manager view with an interactive job scheduler, and a CSRD compliance view with audit-ready carbon reporting — backed by a fully reproducible Python codebase hosted on GitHub. This document records the entire journey: every major decision, the alternatives weighed against it, the methods that worked, those that did not, and the significance of what was found.

2. Introduction and Background

1.1 What is a data centre?

A data centre is a physical building housing hundreds or thousands of computers — servers — that run continuously to deliver digital services: web applications, data storage, artificial-intelligence model training, scientific simulation. When a person streams a video or sends an email, the request is processed by servers in a data centre. Because these servers never switch off, they consume very large amounts of electricity. In 2024 data centres consumed roughly 415 TWh globally, comparable to the annual electricity use of a mid-sized country, and the International Energy Agency projects this could reach 945 TWh by 2030.

1.2 What is high-performance computing?

A high-performance computing (HPC) centre is a specialised data centre built for a single purpose: executing extremely large scientific calculations that no single computer could handle. Thousands of powerful machines are wired together to act as one enormous computer. HPC centres simulate weather and climate, model protein folding for drug discovery, run nuclear-physics simulations and train large AI models. Jobs may run for hours, days or weeks and can occupy thousands of processors at once.

The facility at the centre of this project is **ORNL Summit**, located at Oak Ridge National Laboratory in Tennessee and operated by the U.S. Department of Energy. When built in 2018 it was the world's most powerful supercomputer. Summit comprises **4,626 computing nodes**, each a powerful computer in its own right, and at full load draws approximately 13 megawatts — enough to power around 10,000 homes.

1.3 Why the source of electricity matters

Not all electricity is equal from a climate perspective. Power generated by burning coal, gas or oil releases carbon dioxide; power from nuclear, solar, wind or hydro produces little or none. The measure of how much CO₂ is released per unit of electricity is **carbon intensity**, expressed in grams of CO₂ per kilowatt-hour (gCO₂/kWh). The higher the number, the dirtier the electricity.

The decisive insight behind this project is that carbon intensity is **not constant**. It changes every hour with the mix of power plants currently generating. At 2 a.m. on a windy night a grid might be dominated by nuclear and wind (low intensity); at 7 p.m. on a hot evening every fossil plant may run at full capacity to meet peak demand (high intensity). A computational job that could be deferred a few hours to a cleaner window would emit substantially less CO₂ for exactly the same computation.

1.4 The opportunity

Many HPC and data-centre jobs are time-flexible. A scientific parameter sweep due 'sometime this week', a nightly backup, a machine-learning training run or a batch report does not need to start immediately. If such delay-tolerant work could be shifted into the cleanest hours, an organisation would cut its Scope 2 emissions without buying any new hardware and without performing less computation. This project quantifies that opportunity using real operational data, and builds the tools to capture it.

1.5 Regulatory context

The urgency is amplified by regulation. The EU Corporate Sustainability Reporting Directive (CSRD), effective from 2025, requires large organisations to disclose Scope 1, 2 and 3 greenhouse-gas emissions with third-party assurance. For data centres, Scope 2 emissions from purchased electricity are the dominant contributor. The European Energy Efficiency Directive further requires data centres above 2,780 MWh annual consumption to report performance metrics including Carbon Usage Effectiveness. A system that both reduces and credibly reports Scope 2 emissions therefore addresses a live commercial and compliance need.

3. Problem Statement and Objectives

2.1 Problem statement

Data centres and HPC facilities consume electricity around the clock, yet the carbon footprint of that electricity changes significantly every hour depending on how it is generated. Most facilities schedule workloads on computational priority alone, with no awareness of these fluctuations, so intensive jobs routinely run during the dirtiest hours of the day simply because a server became available.

This project addresses that gap by forecasting grid carbon intensity 48 hours ahead and recommending optimal scheduling windows for flexible workloads, reducing Scope 2 emissions without reducing computation. While ORNL Summit is used as the data source — being the only publicly available high-resolution per-node power dataset from a major operational computing facility — the methodology is directly replicable for any in-premise data centre running delay-tolerant workloads such as batch jobs, backups and model training.

2.2 Why ORNL, and why this generalises

ORNL Summit is an extreme example of a computing facility, but it was chosen as a data source for a deliberate reason: it is the only public dataset providing per-node, per-minute power telemetry from a major operational facility, and it is peer-reviewed. The scheduling mechanism — identify which jobs can wait, forecast when the grid will be cleanest, run them then — is identical for any facility. In a typical corporate data centre the delay-tolerant workloads are nightly database backups, batch ETL and reporting pipelines, machine-learning training, video transcoding, log aggregation and software builds. Indeed, because HPC scientific jobs tend to be more tightly constrained than corporate batch work, the flexible fraction assumed here is a conservative lower bound for most commercial settings.

2.3 Objectives

- Acquire and validate real demand (facility power) and supply (grid generation) data from authoritative sources.
- Integrate the two into a single hourly dataset and compute grid carbon intensity using internationally recognised emission factors.
- Characterise, through descriptive analysis, when the facility consumes power and when the grid is cleanest — establishing whether a scheduling opportunity genuinely exists.
- Forecast grid carbon intensity 48 hours ahead, comparing multiple modelling approaches rigorously on held-out data.
- Optimise the scheduling of flexible workloads into low-carbon windows under realistic node-capacity and deadline constraints, and honestly quantify the achievable saving.
- Deliver the results through an interactive dashboard serving two stakeholders — facility operators and sustainability/compliance officers — and deploy it publicly at no cost.

2.4 Alignment with the Sustainable Development Goals

The work aligns directly with SDG 13 (Climate Action) by reducing greenhouse-gas emissions from computing, and with SDG 7 (Affordable and Clean Energy) by increasing the effective utilisation of clean generation that would otherwise be displaced by fossil dispatch at peak hours.

4. Literature Review

3.1 The scale of the problem

Data-centre electricity demand is one of the few sectors where emissions are projected to grow rather than decline (IEA, 2025). The challenge is not merely the magnitude of consumption but that it is drawn from grids whose carbon intensity fluctuates hourly. Most facilities schedule without awareness of these fluctuations, so computationally intensive jobs frequently run during periods of high fossil-fuel dispatch.

3.2 Evidence that carbon-aware scheduling works

The feasibility of carbon-aware scheduling has been demonstrated at three levels of evidence. In production, temporal shifting has been deployed across more than 20 data centres on four continents, where day-ahead carbon-intensity forecasts were used to reshape cluster power profiles, achieving measurable reductions during peak-carbon hours (Radovanovic et al., 2023). In simulation, carbon-aware scheduling tested against HPC workload traces and national grid data has achieved up to 40.5% carbon reduction compared with conventional scheduling, with only an 8% increase in average job wait time, while an energy-only scheduler on the same workload achieved just 14.3% — confirming that carbon-intensity awareness, not merely energy efficiency, drives the reductions (Anasuri and Pappula, 2023). A broader survey of scheduling frameworks finds that greedy deferral heuristics capture over 90% of the achievable savings relative to complex optimal solutions (Emergent Mind, 2025).

It is important to read these headline figures in context. The 40.5% reduction was achieved on grids and workloads with large carbon swings and high scheduling freedom. As this project's own results show, a nuclear-heavy grid such as TVA, combined with real node-capacity constraints, yields far more modest realistic savings. Reporting the achievable figure honestly, rather than citing the most favourable literature value, is a deliberate methodological stance taken throughout.

3.3 The flexible fraction of HPC workloads

No single published study gives an exact split of delay-tolerant versus time-critical HPC jobs, because it varies by facility and scientific domain. The Bag-of-Tasks pattern — independent, parallelisable jobs — accounts for up to 70% of jobs on parallel systems, though these may consume a smaller share of total CPU-hours (Rodrigo et al., 2018). Actual utilisation in HPC clusters often reaches only about 60% of allocated capacity, with roughly 40% of computational units idling at any moment due to over-provisioning and scheduling fragmentation (Vergara Larrea et al., 2024) — a natural buffer within which flexible jobs can be shifted. At NERSC, 82–87% of jobs ran for two hours or less and 60% used under half their requested walltime, indicating significant runtime over-estimation and natural scheduling slack (Rodrigo et al., 2018). Synthesising this evidence, the project adopts a conservative **30% flexible-workload assumption** with a sensitivity range of 20–40%.

3.4 The regulatory imperative

The EU CSRD (European Commission, 2024) mandates Scope 1–3 disclosure with assurance from 2025; for data centres Scope 2 dominates. The methodology in this project uses IPCC AR5 lifecycle emission factors precisely because they are the basis recognised under CSRD and the GHG Protocol, making the carbon figures audit-ready rather than merely indicative.

3.5 Positioning of this contribution

Where existing work has focused on fleet-scale spatial shifting across continents or on theoretical simulations with synthetic workloads, this project is an end-to-end feasibility study that applies temporal shifting to a single on-premise facility using real power telemetry and real grid data, and quantifies the achievable saving under realistic constraints. Its contribution is not a new scheduling algorithm but a rigorous, honest, reproducible pipeline from raw data to a deployed decision-support tool — including the critical, and rarely reported, finding that point-forecast accuracy is the wrong objective for scheduling.

5. Solution Architecture and Methodology

4.1 The analytics lifecycle

The project follows a five-stage analytics lifecycle, each stage feeding the next: Acquisition, Integration, Analysis (descriptive, predictive and prescriptive), and Delivery. This structure mirrors the Data Value Map presented to stakeholders and organises the remainder of this report.

Stage	Purpose	Key technologies
Acquisition	Capture demand and supply data from authoritative sources	pandas, pyarrow, requests, EIA API v2
Integration	Merge, clean, classify and compute carbon intensity	pandas, NumPy
Descriptive	Characterise power and carbon-intensity patterns	matplotlib, seaborn
Predictive	Forecast carbon intensity 48 hours ahead	statsmodels, Prophet, XGBoost, TensorFlow
Prescriptive	Optimise job placement into clean windows	PuLP + CBC solver
Delivery	Present insights to two stakeholder groups	Streamlit, Plotly

A guiding principle throughout is a **data-first mindset**: raw data is never modified; every load is validated at the point of entry; anomalies are treated, not dropped; all constants live in a single configuration file; the pipeline is idempotent; and every output is traceable to its source. These behaviours are revisited in the acquisition and integration chapters.

4.2 Engineering design decisions

Several architectural decisions shaped the codebase and are worth recording explicitly, because they materially affected reproducibility and reliability.

Decision	Choice	Rationale
Configuration	Single settings module	No magic numbers in processing code; one place to audit thresholds, paths and
Raw data	Immutable	Source of truth never overwritten; outputs are regenerated
Storage format	CSV for processed data	Portable, Git-diffable, no database overhead for 120 + 35k row tables
Idempotency	Skip-if-exists	Re-running never corrupts prior results
Module reuse	src/ imported by both pipeline and notebooks	One implementation, used everywhere; no copy-paste drift

4.3 Reproducibility

The complete pipeline runs from a single command, and each notebook re-derives its outputs from the processed data. The repository is public on GitHub, the dashboard is deployed on Streamlit Community Cloud, and the only secret — the EIA API key — is kept out of version control through an environment file. A reader can clone the repository, install dependencies, supply their own free API key and reproduce every figure in this report.

6. Data Acquisition

5.1 How the data is captured

The project draws on **two datasets**, captured in two different ways — one passive, one active. A third input, the IPCC AR5 emission factors, is not a data feed but a set of published reference constants; it is therefore introduced in the Integration chapter (Section 6.2), where it is applied to convert the grid's fuel mix into carbon intensity.

Source 1 — ORNL Summit power telemetry (passive hardware capture)

Summit's own monitoring infrastructure records power continuously during normal operation. Inside each node, two hardware layers emit readings. The NVIDIA Data Center GPU Manager (DCGM) reads the power draw of each of the six V100 GPUs from on-board sensors. Separately, the node's Baseboard Management Controller (BMC) reads the input wattage of each of the two power-supply units (PSUs) — the actual electricity drawn from the facility distribution, physically measured rather than estimated. These readings, averaged to one-minute resolution across all 4,626 nodes, were published as Apache Parquet files on the ORNL Constellation portal, associated with the peer-reviewed paper of Shin et al. (2021). The project downloads five pre-captured snapshot days.

Source 2 — TVA grid generation (active API retrieval)

The Tennessee Valley Authority reports its hourly generation by fuel type to the EIA under FERC Order 830 — a legal mandate, not a voluntary submission. The project actively retrieves this through the EIA Open Data API v2, paginating in 5,000-record pages until the full 2019–2022 range is collected. The data arrives in long format (one row per fuel type per hour) and is saved verbatim before any processing.

5.2 Explicit data sources

#	Dataset	Side	Organisation	Access	Volume
1	Per-node power	Demand	ORNL (US DoE)	Parquet download	5 files, ~4 GB
2	TVA generation by fuel	Supply	EIA (federal)	REST API v2	280,503 rows

5.3 Evaluating source reliability

Each source was assessed on authority, data lineage, completeness and consistency.

ORNL Constellation

- **Authority:** a US Department of Energy national laboratory, one of the world's leading HPC institutions.
- **Peer review:** the exact dataset underpins Shin et al. (2021), published at SC '21, the flagship supercomputing conference.
- **Lineage:** readings come directly from hardware sensors (BMC, DCGM), not estimates.
- **Completeness:** about 0.02% of GPU readings are null (transient sensor dropouts).

- **Known limitation:** only five snapshot days exist — a dataset design choice, not a quality defect, but one that shapes how annual figures must be framed.

EIA API

- **Authority:** the primary US federal energy-statistics agency.
- **Legal basis:** data reported under FERC Order 830, a regulatory obligation.
- **Cross-validation:** TVA's published annual generation matches the API totals.
- **Completeness:** 1,160 null values (0.4%), concentrated in solar in early 2019 when TVA had negligible solar capacity.
- **Known anomaly:** occasional negative values (pumped-hydro storage, solar net-metering artefacts) — physically meaningful, handled in integration.

5.4 Options considered and why the chosen approach was selected

For demand data, three alternatives were rejected before settling on ORNL.

Option	Why rejected
Synthetic workload traces	Not representative of real production; a simulation, not an evidence base
NERSC Perlmutter telemetry	Per-node, per-minute PSU power not publicly available at this resolution
Cloud (Google/AWS/Azure) data	Not published at node level; only aggregate sustainability reports exist
ORNL Summit (chosen)	Only public per-node-per-minute PSU telemetry from a major facility; peer-reviewed; PSU input

For supply data, the chosen EIA API was weighed against commercial and regional alternatives.

Option	Why rejected
Electricity Maps API	Paid; historical data behind paywall; not reproducible
WattTime API	Provides marginal (not average) emissions; commercial licence
Carbon Intensity UK API	Wrong geography — ORNL is on the TVA grid
CAISO / MISO feeds	Different balancing authorities; not TVA
EIA API v2 (chosen)	Free, official, TVA-specific, complete for 2019–2022, cross-validated against TVA reports

For emission factors, EPA eGRID and commercial real-time factors were considered but IPCC AR5 was selected because it is the CSRD-recognised basis and applies cleanly to a generation-mix composition rather than to specific plants.

5.5 Data-first behaviours in acquisition

Validation occurs at the point of entry. The acquisition module logs, for every load, the row count, date range, fuel types present, null counts, the number and source of negative values, and the value range. Any deviation is surfaced before processing begins. The figure below, produced during acquisition, validates empirically that the 250-watt threshold used later to classify a node as active

sits in the natural gap of the bimodal GPU-power distribution.

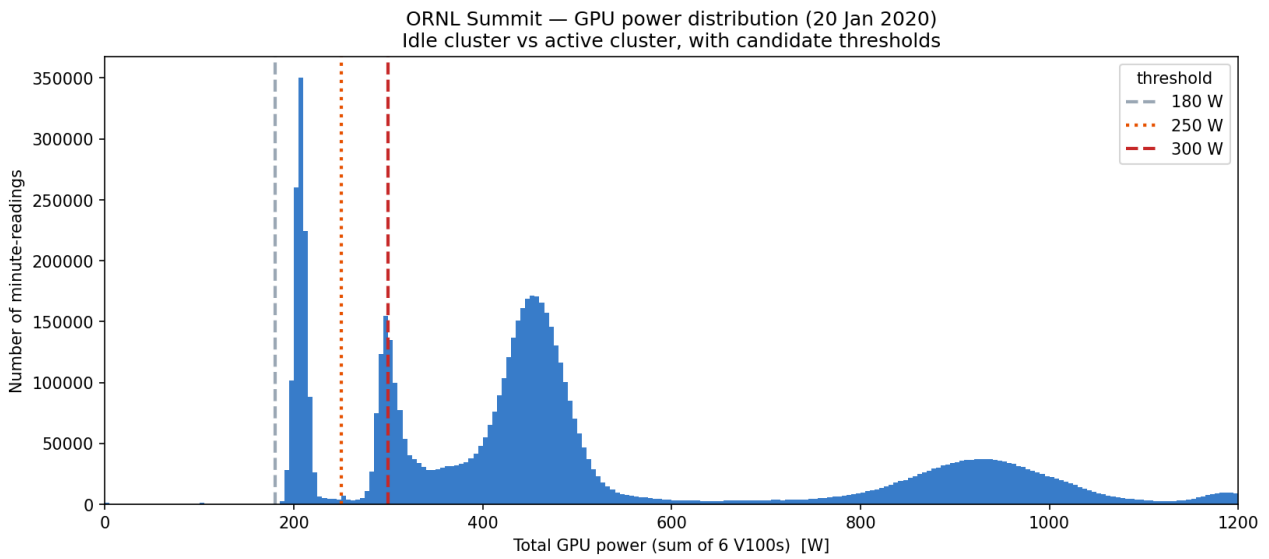


Figure 5.1 Empirical validation of the GPU active-node threshold. The distribution of total GPU power is multi-modal: a dense idle cluster near 200–225 W, a low-density valley around 225–275 W, then the active clusters. The chosen 250 W line (red) sits in the valley; 180 W and 200 W (grey/orange) fall inside the idle cluster and would misclassify idle nodes as active.

The annual generation mix retrieved from the EIA API is shown below, confirming TVA as a nuclear-dominated grid with coal and gas providing flexible capacity.

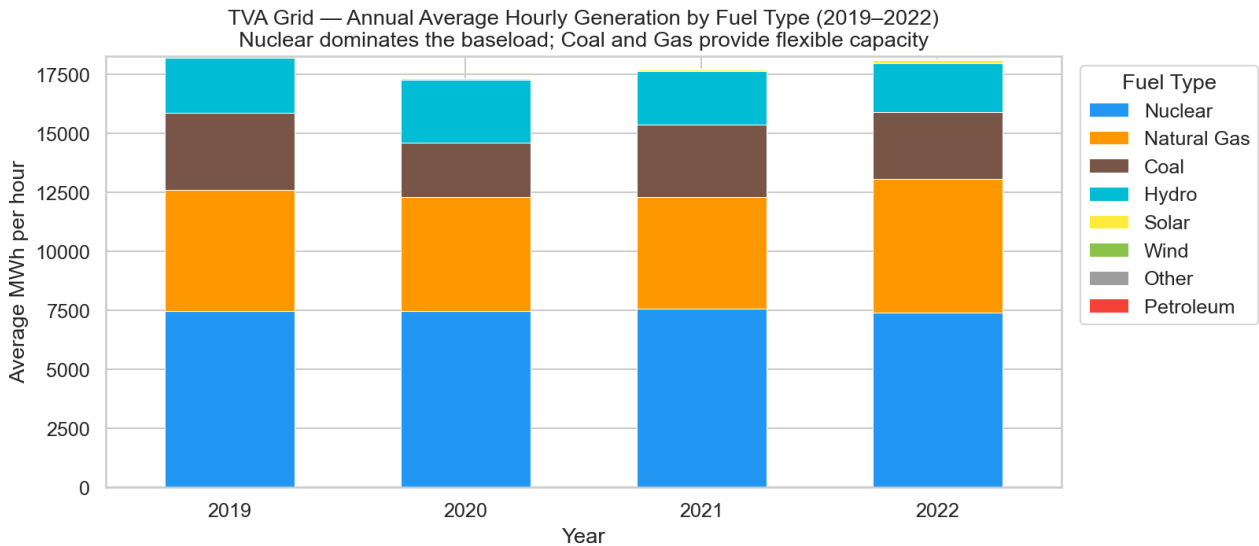


Figure 5.2 TVA annual average hourly generation by fuel type, 2019–2022, retrieved from the EIA API. Nuclear provides the baseload; coal and gas flex to meet demand; wind and solar are negligible on this grid.

7. Data Integration

6.1 Overview

Integration transforms two very different raw sources — minute-level per-node power and hourly grid generation — into a single aligned hourly dataset with carbon metrics. The demand side reduces roughly 34 million raw rows (6.8 million per file × 5 files) into 120 hourly cluster-level rows; the supply side becomes 35,064 hourly rows; and the two are joined into a 120-row integrated table.

6.2 The integration decisions

Twelve decisions defined this stage. Each is recorded with its rationale and the alternative rejected, because integration is where most of the project's methodological risk lived.

Decision 1 — Timezone alignment (the highest-risk step)

ORNL timestamps are in UTC; TVA/EIA data is in Eastern Prevailing Time. Because Summit physically sits in TVA territory, the grid powering it at any moment is TVA's. The demand timestamps are therefore converted from UTC to US/Eastern before the join. Without this conversion every hour would be misaligned by four or five hours (daylight-saving dependent), silently attributing the wrong grid carbon to each workload hour. The correctness of the alignment is confirmed by the join producing **zero unmatched rows out of 120**. The rejected alternative — joining on raw UTC — would have invalidated every carbon figure undetectably.

Decision 2 — Active/idle classification

The dataset's node-state column reads 'Powered On' for every row and is therefore useless for telling whether a node is running a job. Classification is instead inferred from GPU power: a node is active when its total GPU power across six V100s exceeds **250 W**. The threshold was set by inspecting the actual data rather than assuming a value. The per-node total-GPU-power distribution is clearly multi-modal: a dense idle cluster sits at roughly 200–225 W (idle V100s draw a little above their nominal 30 W once driver and power-management overhead are included), followed by a distinct low-density valley around 225–275 W, before the active clusters begin near 290 W and rise well beyond. A threshold of 250 W sits in the middle of that empty valley, cleanly separating the two states. Lower candidates were rejected on the evidence: 180 W or 200 W fall inside the idle cluster itself (classifying ~98–100% of readings as active, which is implausible), whereas 250 W leaves a safe margin above idle fluctuation yet still catches every genuine workload (Figure 5.1). The rejected alternative of using PSU power as the classifier was discarded because PSU draw includes a large fixed node overhead that blurs the active/idle boundary.

Decisions 3–4 — Two-stage aggregation and the granularity decision

Aggregation proceeds in two stages: minute readings are first averaged to a per-node hourly value, then summed across all nodes to a cluster-level hourly total. This preserves the per-node active/idle split before summing and keeps memory bounded by processing one 800 MB file at a time. Because grid carbon intensity is only published hourly, hourly is the finest meaningful join resolution; demand is therefore aggregated *up* to hourly rather than fabricating sub-hourly grid data.

Decision 5 — Carbon-intensity formula and emission factors

This is the step where the grid's fuel mix becomes a single carbon number, and where the IPCC AR5 emission factors introduced in Section 5.1 are applied. For each hour the carbon intensity is the generation-weighted average of the lifecycle emission factors (IPCC Fifth Assessment Report, Working Group III, Annex III), chosen because they are the methodology recognised under the EU CSRD, making the resulting figures audit-ready:

$$CI \text{ (gCO}_2\text{/kWh)} = \text{sum(generation_fuel \times EF_fuel)} / \text{total_generation}$$

Fuel	Factor (gCO ₂ /kWh)	Note
Coal (COL)	1,000	Highest emitter
Petroleum (OIL)	800	Rare for electricity; included for completeness
Natural Gas (NG)	450	Cleaner than coal; methane leakage in lifecycle
Other (OTH)	500	Conservative geothermal/tidal assumption
Nuclear · Hydro · Solar · Wind	0	No combustion; negligible at lifecycle median

Because generation is in MWh and the factors are per kWh, the unit conversion cancels between numerator and denominator, yielding gCO₂/kWh directly. The result ranges from 75.7 to 499.9 gCO₂/kWh across the four years, with a mean of 283.1.

Decisions 6–8 — Anomaly and missing-value handling

Negative generation values are clipped to zero. Each negative has a physical explanation: TVA's Raccoon Mountain pumped-hydro facility consumes power to pump water uphill (appearing as negative generation, down to –1,155 MWh), and solar occasionally reports small negatives from net-metering. A PSU sensor fault in the January 2020 file produced an impossible –1,102 W reading, also clipped, since a power supply cannot push energy back to the grid. Missing values are filled group-wise — GPU nulls within each node, supply nulls within each fuel type — so a coal gap is never filled from nuclear. Crucially, **no rows are ever dropped**; the continuous hourly series the forecaster needs is preserved.

Decision 9 — Flexible fraction with sensitivity

Because the ORNL data contains no job metadata, shiftability cannot be classified per job; a literature-backed 30% flexible fraction is applied, with every carbon column also computed at 20% and 40%. Computing all three in one pass makes the dashboard sensitivity analysis instantaneous.

Decisions 10–12 — Baseline, annualisation and storage

The optimised case initially modelled shifting flexible energy to the day's lowest-CI hour — a greedy upper bound later refined in the optimisation chapter. Annual figures are projected as mean daily saving × 365 and explicitly labelled projections, because only five snapshot days exist. Processed data is stored as CSV, raw data is kept immutable, and the pipeline is idempotent.

6.3 Challenges experienced

- **Timezone alignment** was the most error-prone step; a silent offset would have been undetectable without the zero-unmatched-rows check.

- **Granularity mismatch** forced a deliberate trade-off — aggregating fine-grained demand up to hourly rather than inventing sub-hourly grid data.
- **Physically real anomalies** (pumped hydro to –1,155 MWh) had to be distinguished from sensor faults (PSU –1,102 W); both were clipped but for different reasons.
- **DST-transition gaps** left three isolated hours with missing fuel rows after pivoting, filled with zero for the near-zero fuels affected.
- **Sparse demand** — five snapshot days rather than a continuous record — shaped the honest 'projection' framing of every annual figure.

6.4 Robustness of the hourly active/idle classification

A node can be active for part of an hour and idle for the rest, so classifying it from its hourly-mean GPU power is an approximation of a sub-hourly mix. This was checked directly. On the January 2020 file, of 111,024 node-hours, 68.3% were active for the full hour, 7.8% idle for the full hour, and **23.9% were mixed** (both active and idle minutes). However, recomputing the day's total active energy with an exact per-minute classification changed the total by only **0.8%** versus the hourly-mean method. The distortion is small because most mixed hours are lopsided, over- and under-counts cancel across 4,626 nodes, and boundary cases carry little energy. This 0.8% is far below the uncertainty already carried by the 30% flexibility assumption (a ± 10 -point band), so the hourly-mean classifier is a sound simplification; the exact per-minute method remains available as a refinement.

6.5 The integrated dataset

Integration yields three clean, fully validated, null-free outputs: a 120-row demand table, a 35,064-row supply table, and a 120-row integrated table carrying baseline and optimised carbon at all three flexibility assumptions. These feed every subsequent stage.

8. Descriptive Analysis

Descriptive analysis answers three questions before any modelling: when does the facility consume power, when is the grid cleanest, and do these currently align or misalign? It is organised across the demand, supply and integrated layers.

7.1 Demand — power-consumption profiles

Across the five snapshot days ORNL Summit averaged 76.5% node utilisation, with active (job-driven) energy representing 65–89% of total draw. Utilisation declined over the observation period — from 89% node utilisation in January 2020 to 56% by January 2022 — consistent with Summit being progressively wound down as its exascale successor, Frontier, was commissioned. This declining trend is significant for the project: lower utilisation means more idle headroom, and therefore more room to shift flexible work.

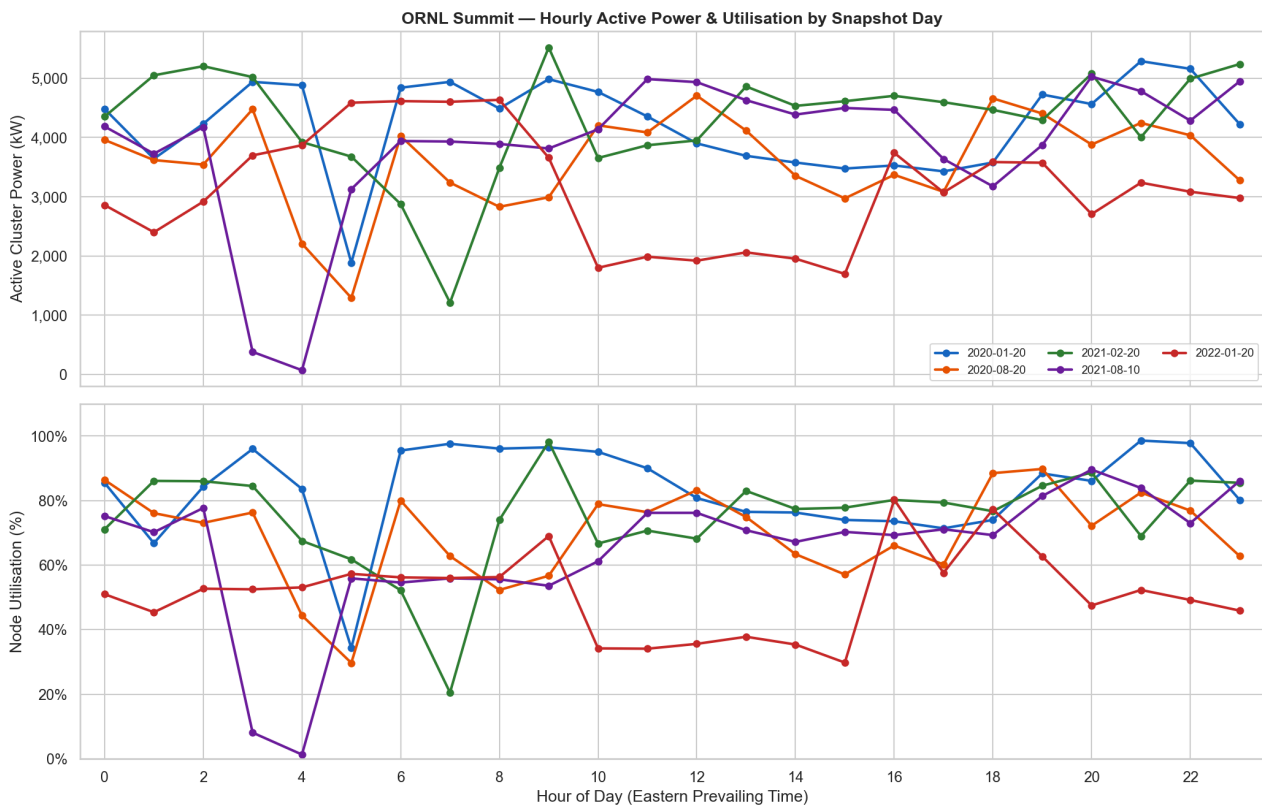


Figure 7.1 Hourly active power and node utilisation for each of the five snapshot days. Utilisation is high but variable within each day, and declines across the two-year span.

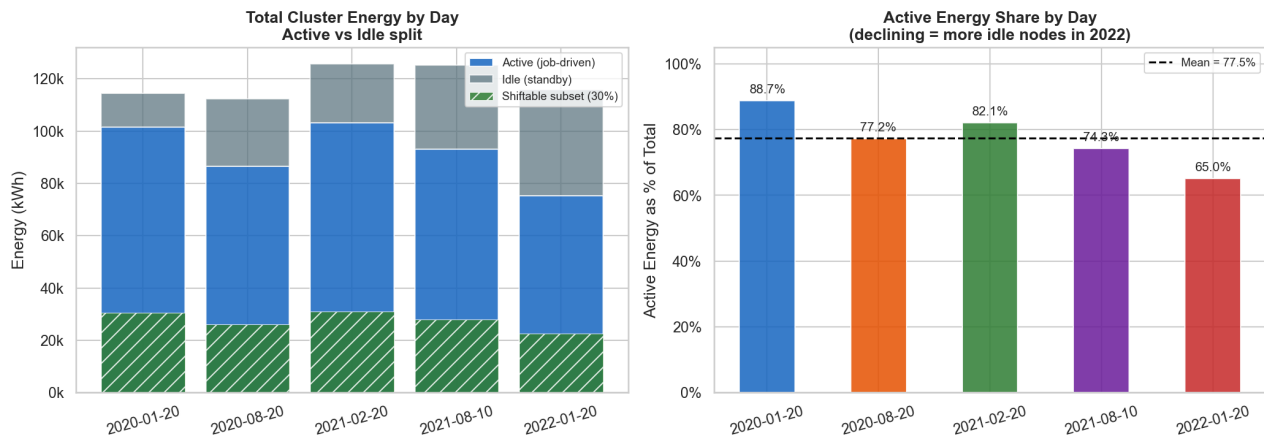


Figure 7.2 Energy split between active (job-driven, shiftable) and idle (standby, fixed) load. Only the active portion can be scheduled; the rising idle share toward 2022 reflects Summit's wind-down.

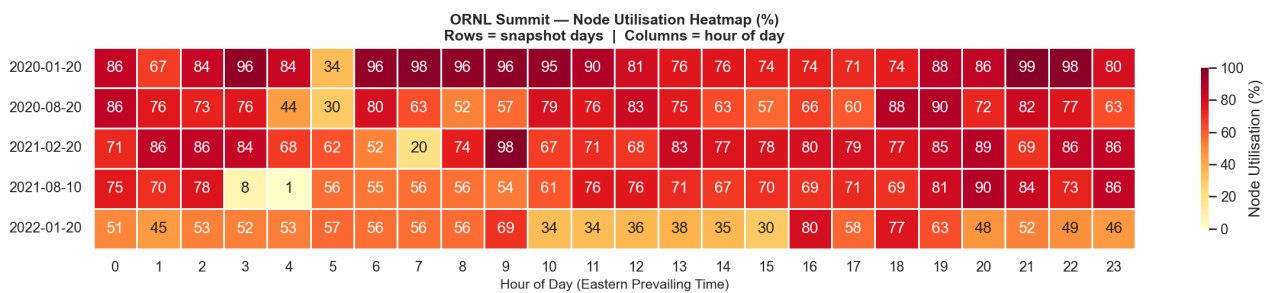


Figure 7.3 Node-utilisation heatmap (five days x twenty-four hours). The absence of a single consistently-idle hour across days foreshadows the central scheduling challenge.

7.2 Supply — grid carbon-intensity patterns

The TVA grid averaged 283 gCO₂/kWh over 2019–2022, ranging from 76 to 500, with a 56.3% low-carbon share dominated by nuclear baseload. The grid decarbonised modestly over the period, from a 2019 mean of 301 to a 2022 mean of 293 gCO₂/kWh.

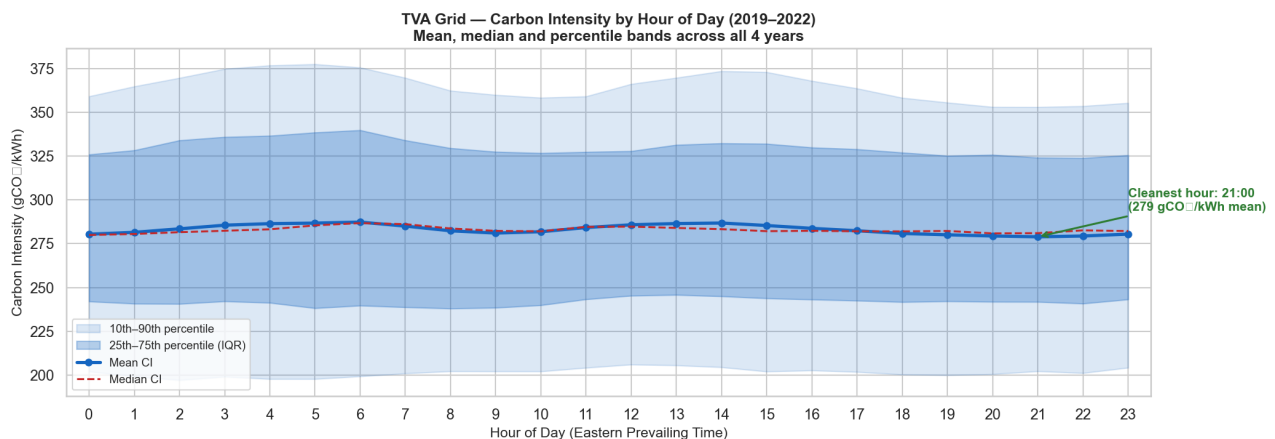


Figure 7.4 Mean carbon intensity by hour of day with percentile bands. Strikingly, averaged across four years the daily profile is nearly flat — the cleanest and dirtiest hours differ by only about 8 gCO₂/kWh.

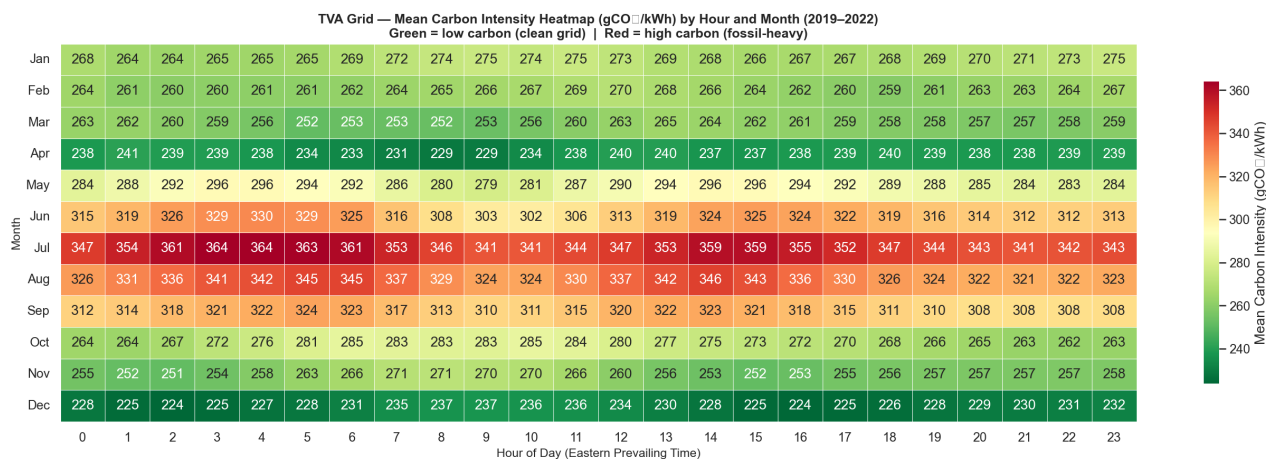


Figure 7.5 Mean carbon intensity by hour and month. Seasonal structure is visible (summer is dirtier), but no fixed hour-of-day is reliably clean across all months.

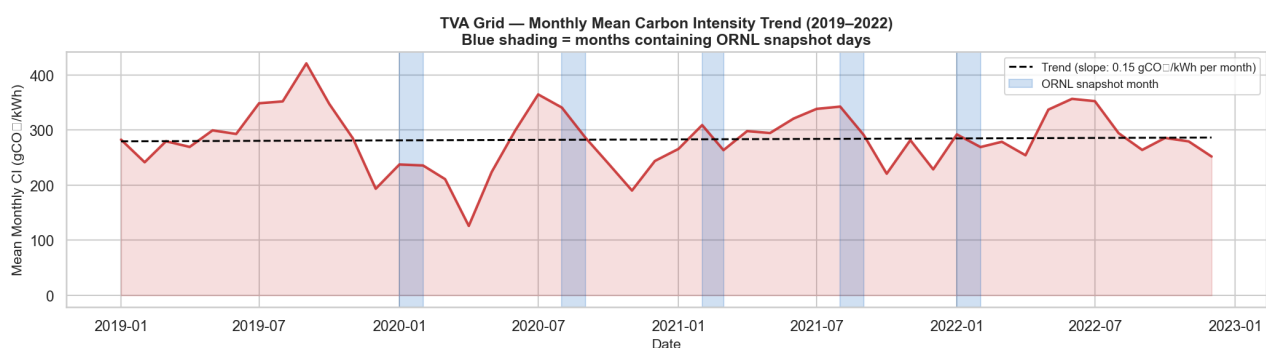


Figure 7.6 Monthly mean carbon-intensity trend, 2019–2022, with the five ORNL snapshot months highlighted. The gentle downward trend strengthens the future case for carbon-aware scheduling.

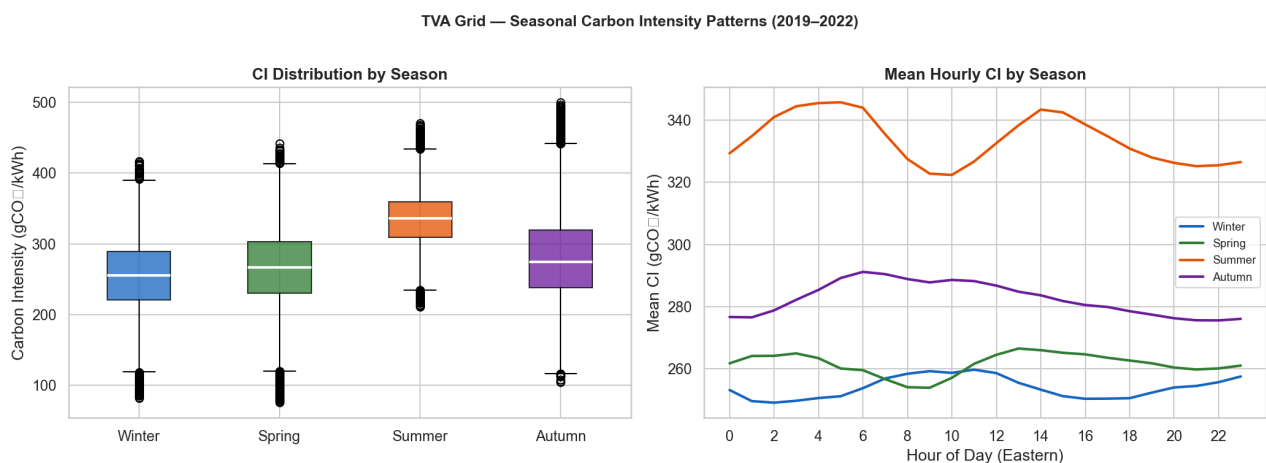


Figure 7.7 Seasonal distribution of carbon intensity. Summer (mean 334 gCO₂/kWh) is markedly dirtier than winter (254), explaining why the August snapshot days saved less carbon than the January ones.

7.3 The central finding — the clean window moves

The single most important result of the descriptive phase emerges from contrasting two facts. Averaged across four years, the hour-of-day spread in carbon intensity is only about 8 gCO₂/kWh (Figure 7.4) — which would suggest time-of-day barely matters. Yet the carbon-intensity range *within*

a single day averages 51 gCO₂/kWh, more than six times larger. The resolution is that on the TVA grid the clean window does not sit at a fixed hour: it moves unpredictably from day to day, driven by nuclear maintenance cycles, hydro availability and demand swings.

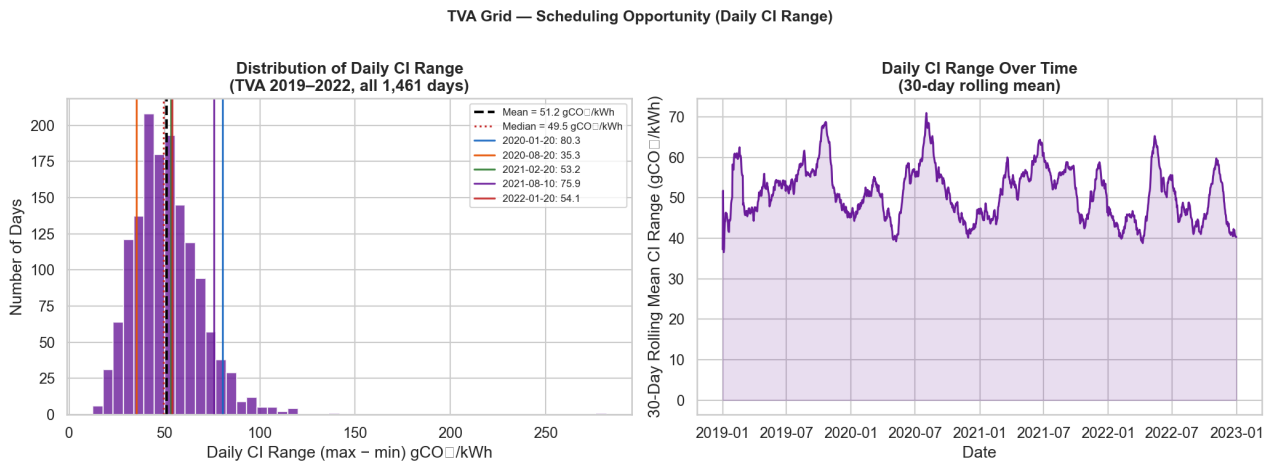


Figure 7.8 Distribution of the daily carbon-intensity range across all 1,461 days. The mean within-day range is 51 gCO₂/kWh and 49% of days exceed 50 — a substantial scheduling opportunity that the flat hourly average conceals.

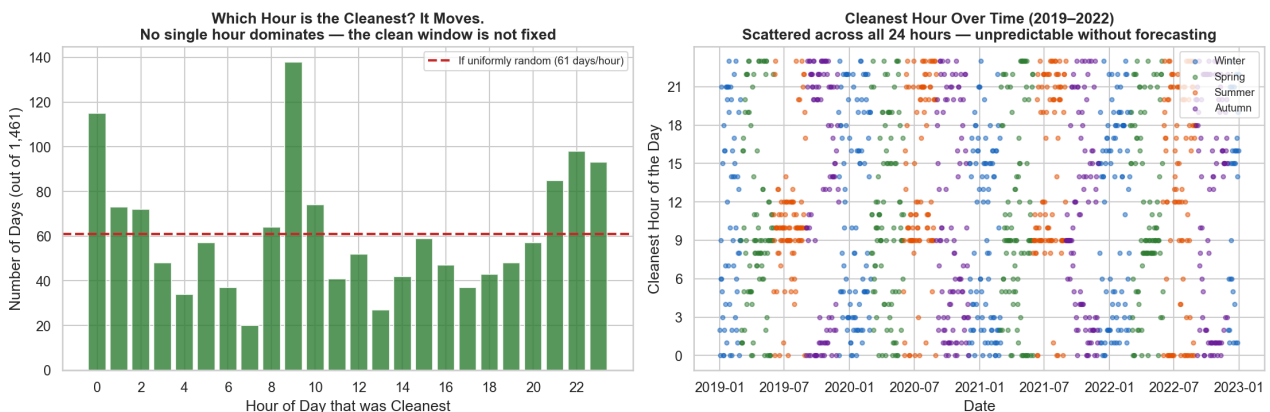


Figure 7.9 The moving clean window. Left: the single most frequently cleanest hour (09:00) is optimal on only 9.4% of days, and every one of the 24 hours is the cleanest on at least ten days. Right: the cleanest hour scattered across the calendar shows no stable pattern.

Because the clean window is not fixed, a static 'always run at hour X' rule would miss it on 90.6% of days. To exploit the 51 gCO₂/kWh daily opportunity, the cleanest window must be forecast for each day. This single finding is the empirical justification for the entire predictive and prescriptive effort that follows.

7.4 Integrated — the opportunity made concrete

Overlaying the grid carbon intensity with the facility's shiftable energy for each observed day visualises the mismatch the scheduler exists to correct.

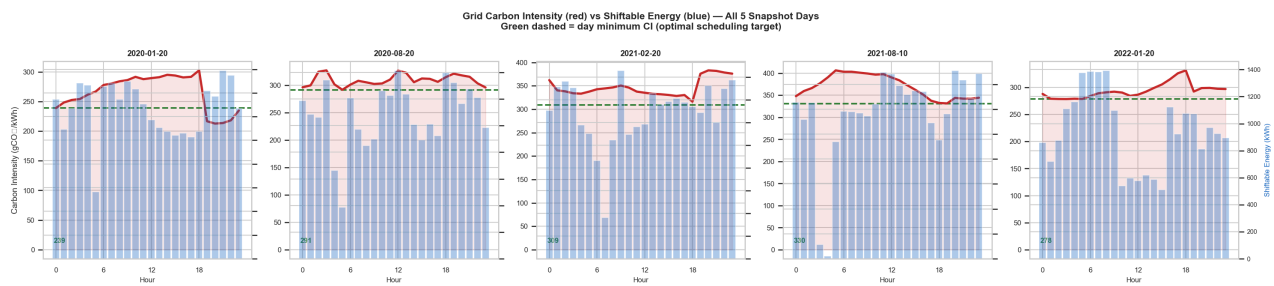


Figure 7.10 Grid carbon intensity (line) against shiftable energy (bars) for each of the five snapshot days, with the day's minimum-CI target marked. The opportunity is the gap between where energy is consumed and where the grid is cleanest.

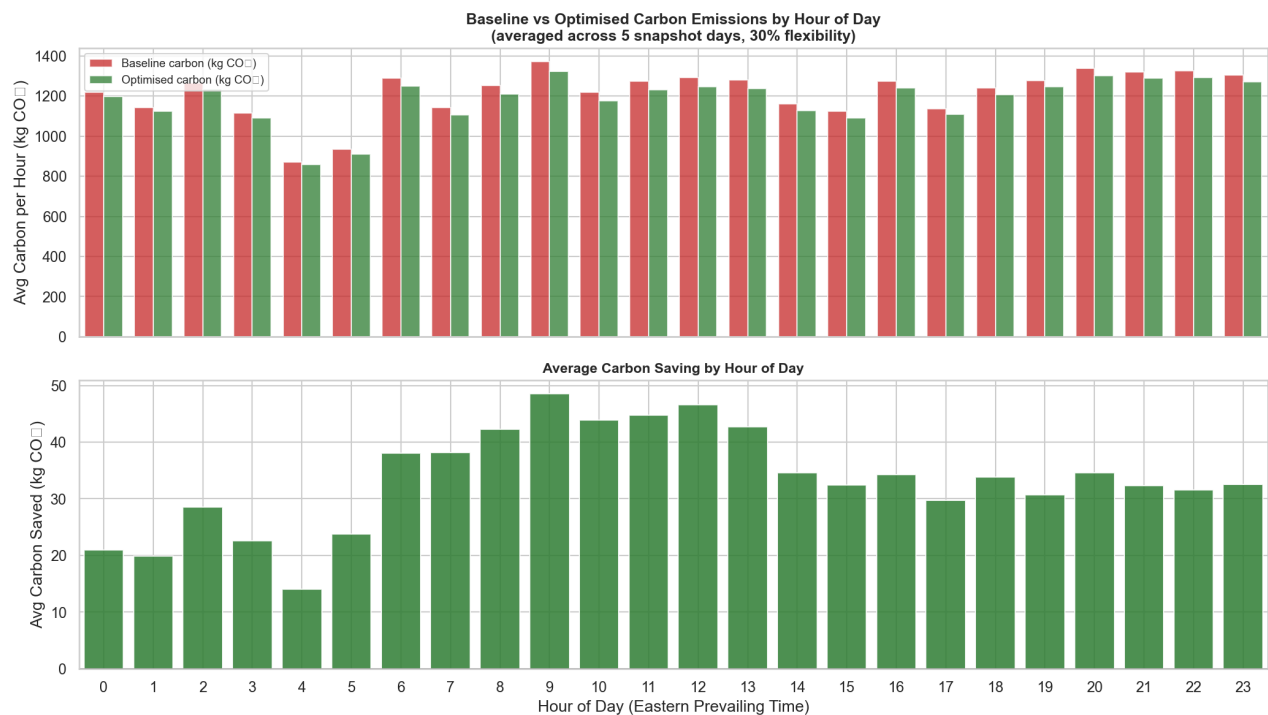


Figure 7.11 Average baseline versus optimised carbon emissions by hour of day, and the carbon saved per hour, indicating where temporal shifting helps most.

9. Predictive Forecasting

8.1 Objective and approach

The descriptive phase established that the clean window must be forecast. The predictive phase builds and rigorously compares models that forecast TVA carbon intensity up to 48 hours ahead. Four models plus a naive baseline were evaluated: a seasonal-naive baseline, SARIMA, Prophet, XGBoost and (deferred) an LSTM.

8.2 Feature engineering

From the supply series, 27 predictor features were engineered: autoregressive lags at 1, 2, 3, 6, 12, 24, 48 and 168 hours; rolling means and standard deviations over 6, 24 and 168 hours; cyclical sine/cosine encodings of hour, day-of-week and month; calendar flags; and 24-hour-lagged fuel-mix shares. Cyclical encoding matters: it tells the model that hour 23 and hour 0 are adjacent on a circle rather than 23 units apart. The first 168 rows, which lack complete lag history, are retained but excluded from training.

8.3 Evaluation methodology

Time-series data cannot be split randomly without leaking the future into the past. The data was split chronologically: training on 2019-01 to 2022-08 (32,136 hours) and testing on the held-out final four months, 2022-09 to 2022-12 (2,928 hours). Models were evaluated by **rolling-origin 48-hour-ahead forecasting**: from origins spaced 24 hours apart across the test set (122 origins), each model forecast the next 48 hours, and errors were pooled into MAE, RMSE and MAPE and broken down by horizon.

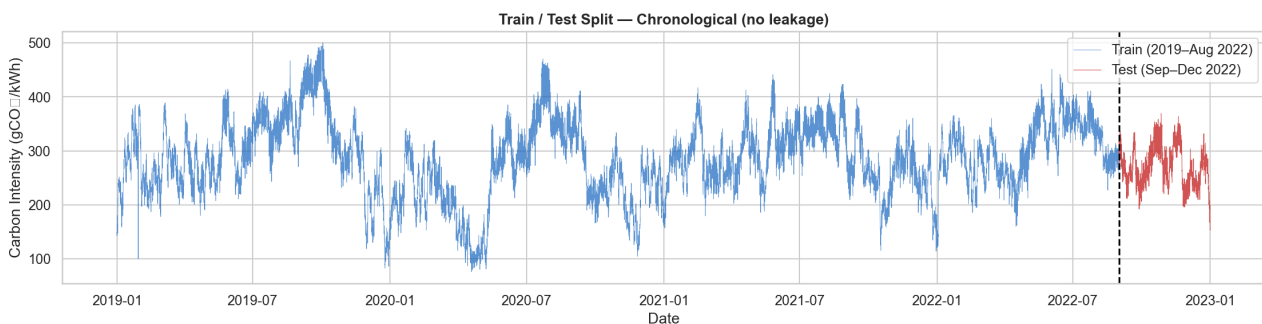


Figure 8.1 Chronological train/test split. Training spans 2019 to August 2022; the held-out test period is September–December 2022. No future information leaks into training.

8.4 The naive baseline — a demanding bar

Before the trained models, two naive baselines were computed. The seasonal-naive 'same hour 24 hours ago' achieved MAE 19.27 gCO₂/kWh (MAPE 6.30%); the weekly version fared far worse at MAE 37.00. The strength of the 24-hour naive reflects the grid's strong daily persistence and sets a demanding bar: any model must beat MAE 19.27 to justify its complexity.

8.5 Model results

Seven models were benchmarked in total: the naive baseline, SARIMA, Prophet and XGBoost, plus three common machine-learning regressors added to test whether XGBoost's win holds up — Linear Regression, Random Forest and k-Nearest Neighbours. All were run through the identical rolling-origin evaluation.

Model	MAE	RMSE	MAPE	Beats naive?
XGBoost (selected)	17.95	23.80	6.80%	Yes
Random Forest	19.16	25.43	7.19%	Yes (just)
Naive (24h)	19.27	25.41	7.35%	— baseline
SARIMA	20.60	26.36	7.95%	No
k-Nearest Neighbours	23.46	29.49	8.78%	No
Prophet	30.39	37.94	11.48%	No
Linear Regression	52.63	68.19	19.75%	No

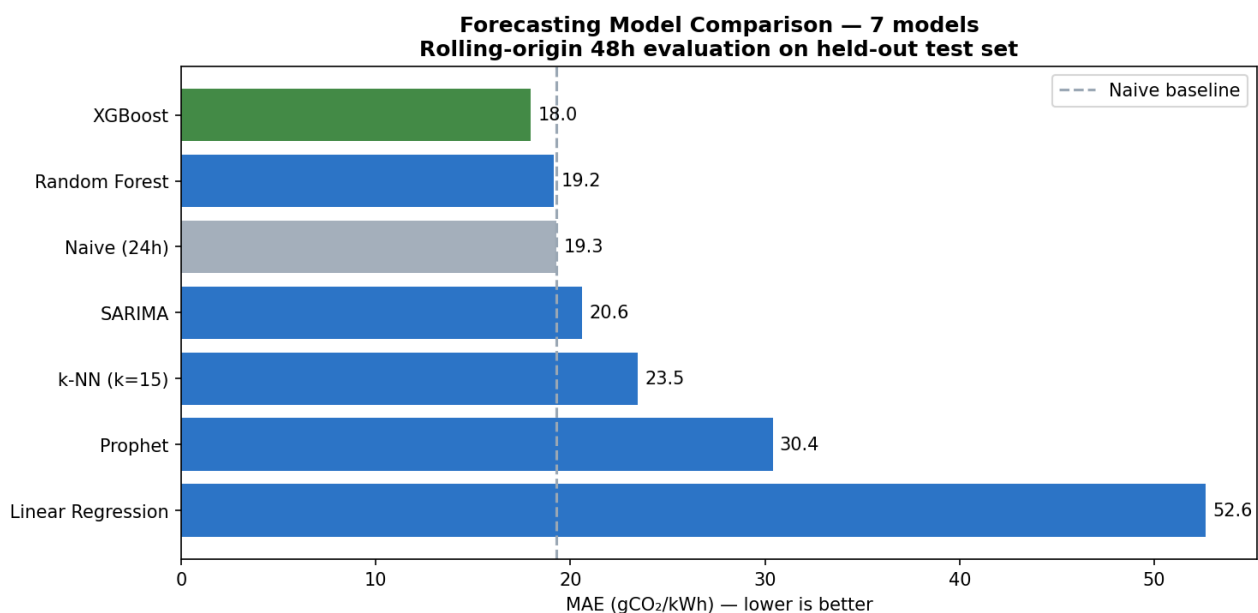


Figure 8.2 Full seven-model comparison (MAE, rolling-origin 48-hour evaluation). Only XGBoost and Random Forest — both tree ensembles — beat the naive baseline.

The extended benchmark reinforces the selection. **Only two of seven models beat the naive baseline, and both are tree ensembles** — confirming that family is the right choice, with gradient boosting (XGBoost) edging out bagging (Random Forest, MAE 19.16). The common alternatives behaved instructively: **Linear Regression failed badly** (MAE 52.63) because, over 48 recursive steps, a linear model extrapolates and compounds its errors; **k-NN was mid-pack** (23.46), hampered by the curse of dimensionality in a 27-feature space. That five of six trained models fail to clear the naive baseline underlines how strongly daily-periodic the TVA grid is.

8.6 What worked, what did not, and why

XGBoost — selected

XGBoost, gradient-boosted trees over the engineered features, achieved the lowest error (MAE 17.95), trained in about two seconds and predicted instantly. Its feature importance confirmed that recent lags dominate — the last hour's carbon intensity alone carries over half the predictive weight.

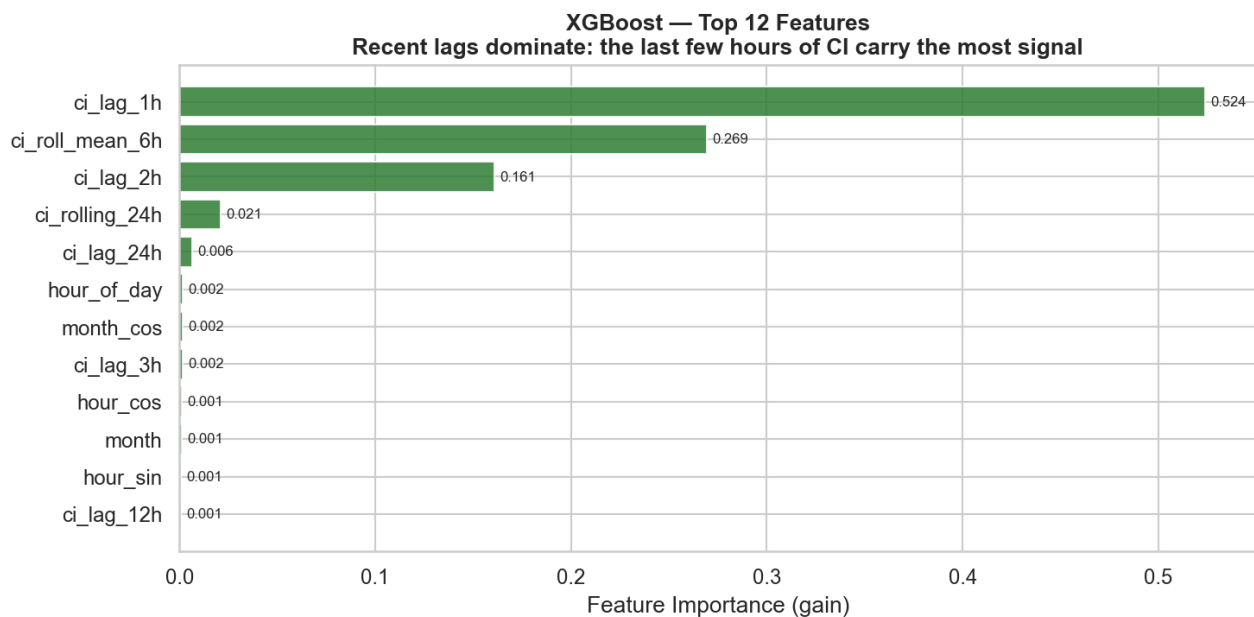


Figure 8.3 XGBoost feature importance. Recent autoregressive lags dominate; calendar and fuel-mix features contribute marginally.

SARIMA — competitive but slower

SARIMA with daily (24-hour) seasonality matched the naive at short horizons but degraded faster and was far slower to fit. To keep it tractable it was fitted on a recent 90-day window and rolled forward with state updates rather than full refits.

Prophet — a corrected near-failure

Prophet initially produced a catastrophic MAE of 122 because, being non-autoregressive, a single global fit ignored recent observations and its trend extrapolation drifted badly over the four-month test horizon. The standard remedy for rolling forecasts — refitting on a trailing 180-day window at each origin, with yearly seasonality disabled and the trend tightened — reduced this to MAE 30.39. It remains the weakest model, an honest reflection that decomposition models are poorly suited to short-horizon, autoregressive grid forecasting.

LSTM — deferred

A stacked LSTM was implemented but proved impractical to train reliably within reasonable time on the available CPU hardware, even after aggressive reduction of sequence length, network size and training-set stride. More importantly, the prescriptive phase later revealed that recursive multi-step neural forecasts suffer the same flattening problem as XGBoost for scheduling, so an LSTM would offer no advantage for the objective that matters. It was therefore deferred and documented honestly rather than forced.

8.7 The horizon finding

Breaking error down by forecast horizon revealed where each model adds value.

Horizon	XGBoost	Naive (24h)	SARIMA	Prophet
h+1 to 6	9.46	15.87	9.93	27.67
h+7 to 24	16.33	16.78	18.15	29.96
h+25 to 48	21.32	22.01	25.15	31.40

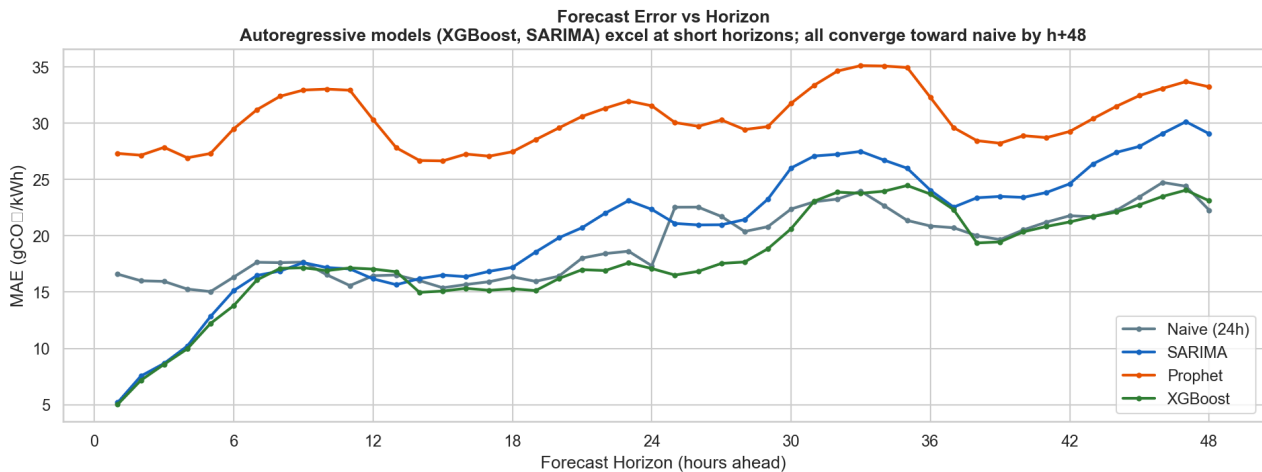


Figure 8.4 Forecast error against horizon. Autoregressive models (XGBoost, SARIMA) excel at short horizons where recent carbon intensity carries signal; by 48 hours all models converge toward the naive, because two days out 'the same as yesterday' is close to the best anyone can do.

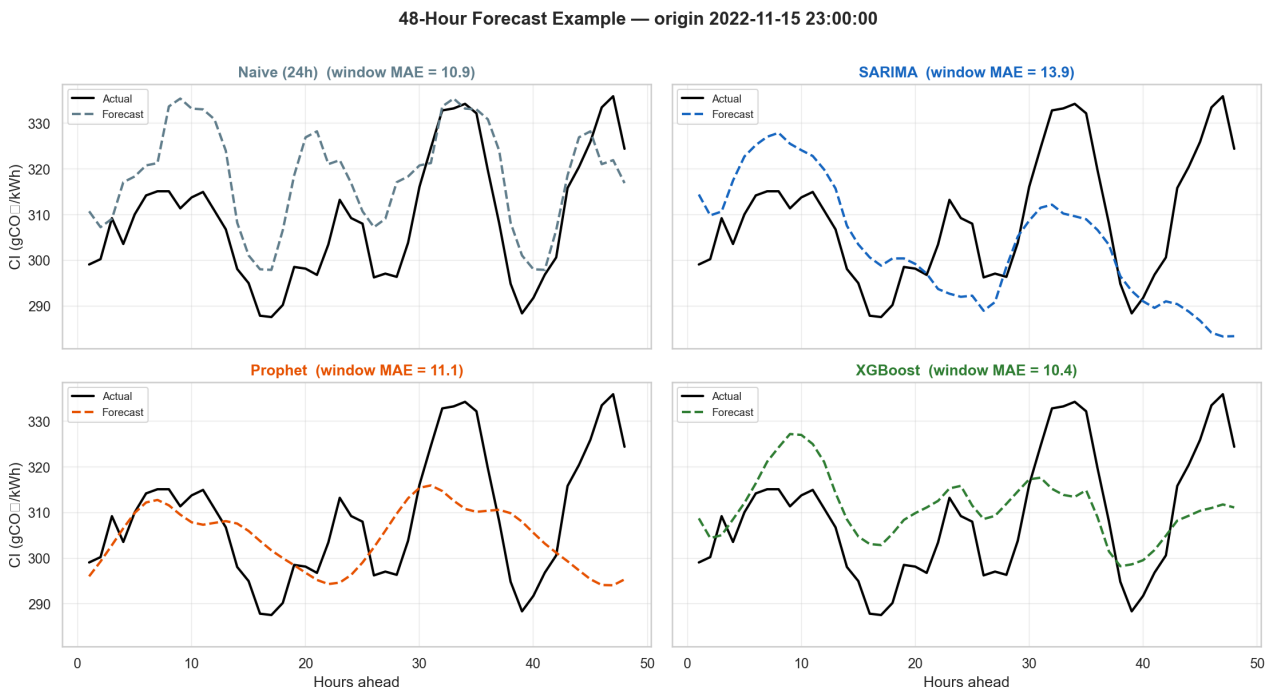


Figure 8.5 Example 48-hour forecasts from each model against the actual carbon intensity for one test window.

The horizon analysis carries a practical implication: a scheduler should weight the near-term forecast most heavily, because that is where the models are genuinely informative. It also foreshadows the prescriptive finding — that point accuracy, which this chapter optimised, is not the same as

scheduling usefulness.

10. Prescriptive Optimisation

9.1 From forecast to decision

The prescriptive phase turns the forecast into an actual scheduling decision: given a job needing a number of nodes for a duration with a deadline, when should it run to minimise carbon? Three schedulers were built — a greedy heuristic, a Linear Programme, and a carbon-blind baseline — plus a fleet-level load-shifting Linear Programme for the headline figure.

9.2 The greedy job scheduler

For a single flexible job the greedy scheduler slides a duration-wide window across the forecast within the deadline and selects the block with the lowest mean carbon intensity that fits node capacity; urgent jobs run immediately. It is fast, intuitive and powers the dashboard's interactive scheduler widget. The literature indicates greedy deferral captures over 90% of achievable savings, making it a strong baseline for the LP.

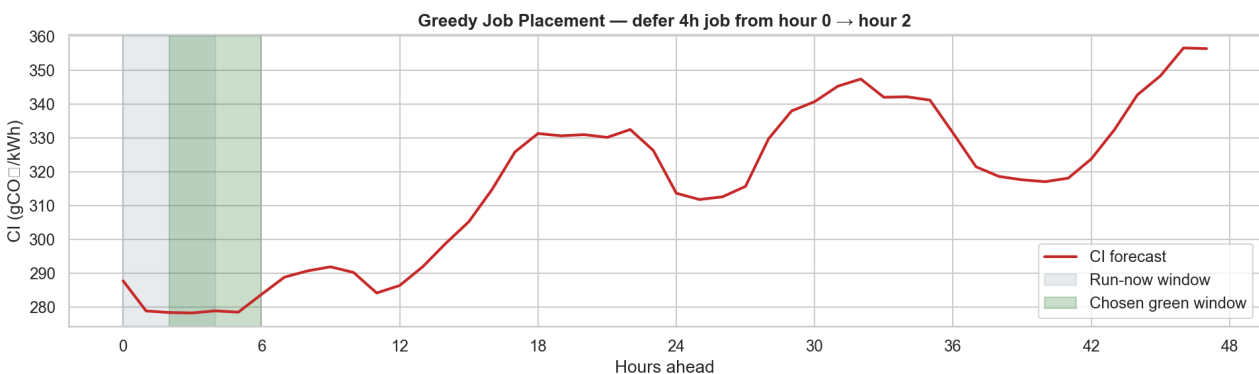


Figure 9.1 Greedy placement of a four-hour job: deferring from hour 0 into a cleaner window later in the forecast reduces its carbon for identical computation.

9.3 The Linear Programme and the correct baseline

When many jobs compete for the 4,626 nodes, greedy's first-come-first-served placement can be beaten by a Mixed-Integer Programme that places all jobs simultaneously, minimising total carbon subject to a one-run-per-job constraint, per-hour node-capacity limits and deadlines, solved with the open-source CBC solver via PuLP. An early test exposed a subtle but important error: comparing against a hypothetical 'everything runs at hour 0' baseline produced *negative* savings, because five jobs totalling 9,800 nodes cannot all run at once on a 4,626-node cluster. The correct as-is baseline is a **carbon-blind scheduler** that packs jobs as early as capacity allows while ignoring carbon — exactly what a real scheduler does today. Against that baseline the LP consistently beats greedy by resolving capacity contention globally, and neither can be worse than the baseline.

9.4 The fleet load-shifting Linear Programme

An important honesty point governs the fleet-level figure. The historical power data contains no job metadata — no job identifiers, deadlines or durations — so the system cannot identify or move individual jobs. Instead it treats each day's active energy as a single **divisible pool** and asks how

much carbon would be saved if the flexible portion (the literature-backed 30%) were shifted from dirty hours to clean ones. This is a load-redistribution model, not a job scheduler, and the 30% assumption is the bridge over the missing job data — which is why the fleet figures are feasibility estimates, presented as a range, rather than a record of jobs actually moved.

Mechanically, a transportation LP redistributes each day's shiftable energy across hours, minimising carbon under the decision forecast and scored against the actual carbon intensity, subject to per-hour capacity headroom and a deferral-only constraint. This corrects the Phase-1 integrated figure, which had implicitly assumed all shiftable energy could be dumped into the single cleanest hour — ignoring that clean hours have limited spare capacity.

9.5 The savings hierarchy

The result is a hierarchy of honest savings figures, each level adding a real-world constraint.

Level	Annual saving	What it accounts for
Unconstrained potential	314 tCO ₂ /yr	Shift to cleanest hour; ignores capacity
Capacity-aware ceiling	87 tCO ₂ /yr	Real node limits; perfect (hindsight) forecast
Realistic forecast-driven	34 tCO ₂ /yr	Capacity limits plus day-ahead forecast error

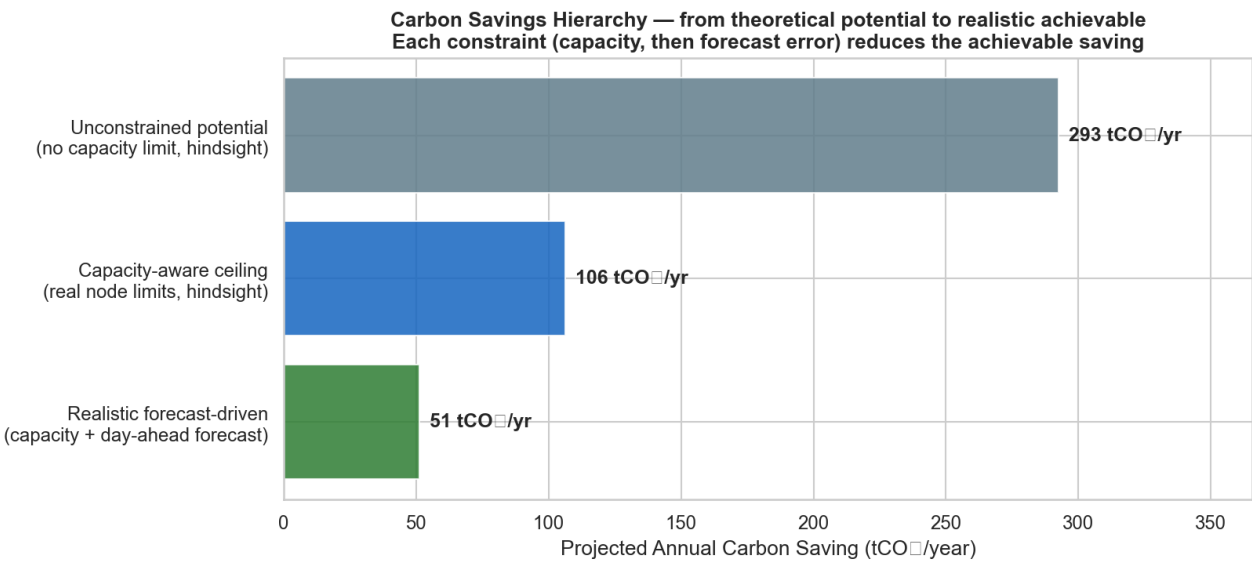


Figure 9.2 The carbon-savings hierarchy. Each constraint — first capacity, then forecast error — reduces the achievable saving from the optimistic potential to a realistic figure. Presenting all three is more defensible than a single headline.

Capacity is by far the largest single constraint: it cuts the achievable saving to only about a quarter of the unconstrained figure (314 → 87 tCO₂/yr). Although roughly half the available flexible energy can be moved, it cannot all reach the very cleanest hours — those hours are already carrying fixed load and fill up quickly — so much of it lands in merely-cleaner hours that save less per kilowatt-hour. This is a genuine and important correction to the optimistic Phase-1 figure, and it is more pronounced at the 250 W threshold, where higher measured utilisation leaves even less idle headroom to absorb shifted load.

9.6 The key finding — forecast shape beats point accuracy

The most significant prescriptive result concerns which forecast to use. XGBoost won the predictive comparison on point accuracy, yet when used to drive the scheduler it performed worst of all — often producing *negative* savings. The reason is that its recursive multi-step forecast reverts to a near-flat line: it captures the average level well but loses the daily *shape*, and a scheduler fed a flat forecast cannot tell clean hours from dirty ones, so it shifts energy to the wrong hours. A simple shape-preserving forecast — yesterday's profile — captured roughly 39% of the capacity-aware ceiling, far more than XGBoost, and it is this shape-preserving forecast, not XGBoost, that drives the realistic saving reported above.

Forecast driving the scheduler	5-day saving	Share of ceiling
Hindsight (perfect CI)	1,198 kg	100% (ceiling)
Yesterday's profile (shape-preserving)	464 kg	39%
XGBoost (best point accuracy)	negative	harmful

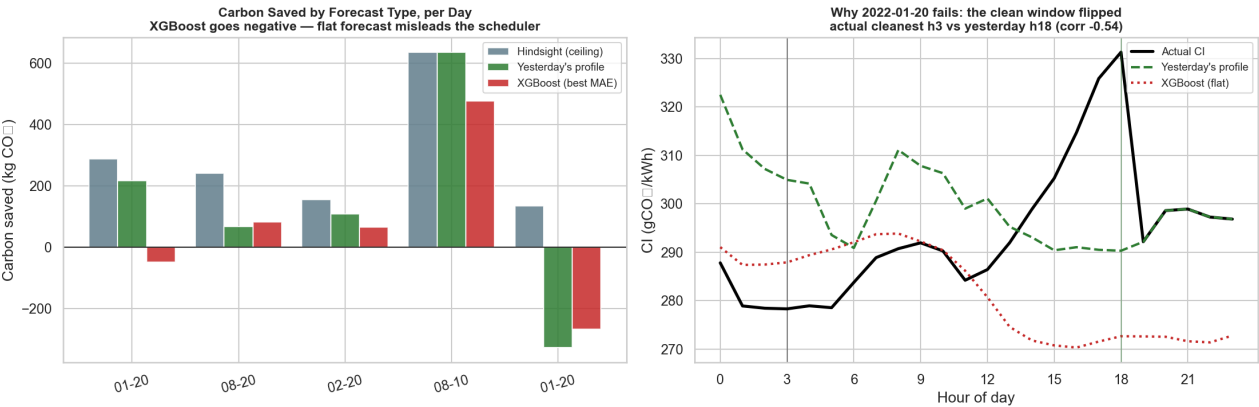


Figure 9.3 Forecast shape matters. Left: carbon saved per day by forecast type — XGBoost goes negative because its flat forecast misleads the scheduler. Right: the 20 January 2022 case, where the actual cleanest hour (03:00) and the previous day's (18:00) were anti-correlated ($r = -0.54$) — the clean window flipped overnight.

The metric that selects a forecaster — point error — is not aligned with the objective that matters for scheduling: the correct ordering of clean and dirty hours. This is a genuine, and rarely reported, insight with direct practical consequences for anyone building such a system.

9.7 Phase 2 and Phase 4 lock together

The catastrophic failure on 20 January 2022 is not a bug but the descriptive 'moving clean window' finding made concrete. On that day the clean window flipped from the previous day (correlation -0.54), so any day-ahead forecast misled the scheduler and committing energy to the forecast-clean hours actively increased carbon. The opportunity is real — the hindsight ceiling proves it — but capturing it is hard precisely because, on this grid, the target moves unpredictably. The descriptive phase predicted this difficulty; the prescriptive phase quantified it. That internal coherence is one of the project's stronger features.

11. Delivery and Deployment

10.1 Choosing the delivery platform

The insights are delivered through an interactive Streamlit dashboard. Streamlit was chosen over Power BI and Tableau because the dashboard integrates directly with the Python analytics pipeline — the same environment used for forecasting and optimisation. This eliminates any need to export model outputs to an external tool and allows the scheduler to run live within the application.

Option	Why rejected / chosen
Power BI / Tableau	Would require exporting model outputs to an external tool, breaking the Python pipeline; no native way
Static report / slides	No interactivity; cannot run the scheduler against user input
Streamlit (chosen)	Runs the analytics pipeline in-app; free public hosting; fast to build; native Python

10.2 Two stakeholder views

The dashboard serves two decision-makers with distinct needs.

Operations Manager view

- Headline tiles: current grid carbon intensity, forecasted low, active node count, mean utilisation.
- A 48-hour forecast chart overlaying carbon intensity and electricity price on twin axes.
- An interactive job scheduler: the operator enters node count, duration, deadline and priority, sets a **carbon-vs-cost balance** on a slider, and the optimiser returns the recommended start time together with both the carbon saved (kg CO₂) and the cost saved (\$) in real time.

CSRD Compliance view

- Headline tiles: projected annual saving, Scope 2 reduction, average grid carbon intensity, low-carbon share.
- The savings hierarchy, baseline-versus-optimised emissions, the grid fuel mix and the monthly carbon-intensity trend.
- An audit-ready CSV export and a methodology note citing IPCC AR5 factors and the flexibility assumptions — making the output suitable for CSRD Scope 2 disclosure.

10.3 Engineering for free, reliable deployment

The dashboard was deliberately engineered to deploy free on Streamlit Community Cloud. It reads small precomputed files — about 50 kilobytes of CSV and JSON — and runs a numpy-only greedy scheduler, so it needs none of the heavy machine-learning libraries at runtime. The runtime dependency set is therefore just four packages (Streamlit, pandas, NumPy, Plotly), kept separate from the full pipeline dependencies. This makes the cloud build fast and robust. The repository is public on GitHub and the only secret, the EIA API key, is excluded from version control.

10.4 Technology decisions summary

The architecture spans six layers, each with a deliberate technology choice: Parquet and the EIA REST API for sources; pandas, pyarrow and requests for ingestion; pandas and NumPy for integration; matplotlib, statsmodels, Prophet, XGBoost and PuLP for analytics; and Streamlit with Plotly for delivery — all coordinated by a single configuration module, structured logging and Git version control.

12. Consolidated Findings and Insights

11.1 The findings, and why they matter

Finding 1 — The clean window moves unpredictably

On the TVA grid the cleanest hour of the day moves from day to day; the most frequent clean hour is optimal only 9.4% of the time. **Significance:** this rules out simple static scheduling rules and justifies the whole forecasting effort. It also explains why savings on a nuclear-heavy, low-variability grid are inherently harder to capture than the headline figures from solar-heavy grids in the literature.

Finding 2 — Point-forecast accuracy is the wrong objective for scheduling

XGBoost minimised forecast error but failed at scheduling because its recursive forecast lost the daily shape; a shape-preserving forecast did far better. **Significance:** this is a transferable lesson for any carbon-aware scheduling system — models should be selected and evaluated on the ordering of clean hours, not on aggregate error. It is a subtle point rarely made explicit in the literature.

Finding 3 — Real constraints sharply reduce achievable savings

Capacity constraints roughly halved the achievable saving, and forecast error roughly halved it again: 314 → 87 → 34 tCO₂/year. **Significance:** honest, constraint-aware estimation produces a credible business case rather than an inflated one, and identifies cluster headroom and forecast quality as the two levers that most increase savings.

Finding 4 — The naive baseline is hard to beat

The 'same hour yesterday' baseline achieved MAE 19.27, beaten only modestly by XGBoost. **Significance:** grid carbon intensity is strongly daily-periodic; sophisticated models add most value at short horizons, and a system should not assume complexity is automatically warranted.

Finding 5 — Utilisation headroom is rising

Summit's node utilisation fell from 89% in early 2020 to 56% by 2022. **Significance:** falling utilisation increases the idle headroom available for shifting, so the opportunity grows precisely as facilities are under-utilised — a favourable dynamic for adoption.

11.2 How the findings interlock

The findings are not independent observations but a coherent chain. The moving clean window (Finding 1) makes day-ahead shape hard to predict, which is why point-accurate but shape-poor models fail at scheduling (Finding 2), which in turn is a major reason the realistic figure falls so far below the capacity-aware ceiling (Finding 3). The project's descriptive phase predicted a difficulty that its prescriptive phase then measured — an internal consistency that lends the conclusions weight.

13. Business Value

12.1 Value to two stakeholders

The Operations Manager uses the scheduling dashboard to shift flexible workloads into low-carbon windows, projecting a realistic 34 tonnes of CO₂ per year of avoided Scope 2 emissions from ORNL Summit alone, at zero additional hardware cost and with no reduction in computation. The Sustainability and Compliance Officer uses the CSRD view to track savings and export audit-ready reports, meeting EU CSRD Scope 2 disclosure requirements.

12.2 Why the value proposition is strong despite modest tonnage

- **Zero capital expenditure** — it is a software change to the scheduling layer, not new infrastructure.
- **No performance penalty** — the same computation is performed; only its timing changes.
- **Compliance-ready** — hourly Scope 2 data and IPCC-based reporting directly serve CSRD disclosure, which is otherwise costly to produce.
- **Replicable and scalable** — the same method applies to any in-premise data centre with delay-tolerant workloads, where the flexible fraction is typically higher than in HPC, so the per-facility saving would be larger.
- **Improving over time** — as grids decarbonise and renewable variability grows, both the clean-window depth and the value of shifting increase.

12.3 The honest framing as an asset

Presenting the full savings hierarchy rather than a single optimistic number is itself a business asset: it gives a sustainability officer a defensible figure that will survive third-party assurance, and gives an operations manager a clear understanding of the two levers — cluster headroom and forecast quality — that would increase the saving.

12.4 The cost dimension — joint carbon and cost optimisation

The project's primary metric is carbon, but the scheduler was extended to co-optimize **electricity cost** alongside it, so the value proposition is not limited to emissions. The Operations Manager view carries a slider that sets a carbon-versus-cost balance; the optimiser blends the (normalised) carbon-intensity and electricity-price forecasts and returns the recommended start time with both the carbon saved and the money saved. Because TVA is a regulated utility with no public hourly market price, the cost side uses a representative Time-of-Use tariff — cheap overnight, expensive on-peak — clearly labelled as a model that a real deployment would replace with the facility's actual contract.

An important honesty point governs this feature: **carbon-optimal is not the same as cost-optimal**. Clean hours are driven by the generation mix while cheap hours are driven by demand; they often overlap (overnight tends to be both cleaner and cheaper) but not always. On an evening-peak job, for example, deferring overnight can save both carbon and cost simultaneously, whereas chasing the single cleanest hour may push the job into an expensive on-peak slot. The slider makes this trade-off

explicit and lets the operator choose — prioritise carbon, prioritise cost, or balance the two — rather than hiding it. This turns the tool from a carbon-only utility into a genuine operational decision aid, while the honest caveat keeps it from over-claiming an automatic bill reduction.

12.5 Cost–carbon trade-off and ROI

To quantify the financial dimension, the load-shifting optimiser was swept across the full carbon-versus-cost weighting, recording for each setting the annual carbon saved and the annual electricity cost saved. This traces a Pareto trade-off frontier (Figure 12.1), with carbon then monetised at roughly \$85/tCO₂ (EU ETS level) to give a single annual benefit.

Setting	Carbon (tCO ₂ /yr)	Electricity (\$/yr)	Total benefit (\$/yr)
Pure cost (w=0)	23.9	\$71,710	\$73,742
Balanced (w=0.5)	57.7	\$64,008	\$68,917
Pure carbon (w=1)	87.5	–\$4,837	\$2,597

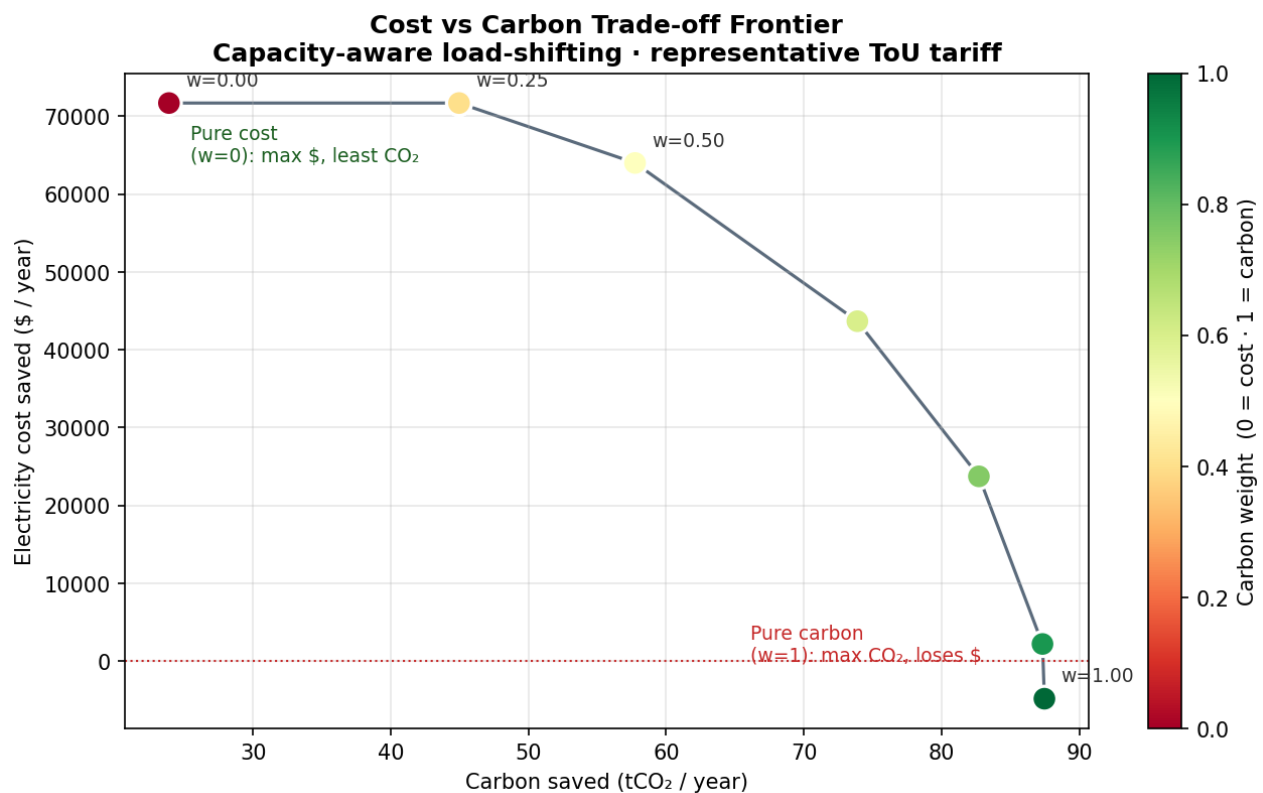


Figure 12.1 Cost-versus-carbon trade-off frontier. As the weighting moves from cost to carbon, the recommended schedule trades electricity savings for emissions savings; the flat left region is a 'free-carbon' zone where both are captured at once.

Three findings stand out. First, in money terms the **electricity cost savings (up to ~\$72,000/yr) dwarf the monetised carbon (~\$2,000–7,400/yr) by a factor of ten to twenty-five**: the financial case rests on the bill, with carbon reduction as a valuable co-benefit. Second, there is a **'free-carbon' region** — at low carbon weight the full cost saving is captured together with about 45 tCO₂/yr, because cheap off-peak hours are often also cleaner. Third, **chasing maximum carbon can cost money** (pure carbon loses ~\$4,800/yr on the bill by pushing jobs into expensive on-peak hours),

which is exactly the trade-off the slider exposes. Because the intervention is a software change costing essentially nothing, an annual benefit of tens of thousands of dollars implies effectively immediate payback — carbon-aware scheduling here is a **negative-cost abatement measure**, reducing emissions while being paid to do so. A structural bonus is that the ToU tariff is known in advance with no forecast error, so the cost savings are more reliably capturable than the forecast-dependent carbon savings. The dollar magnitudes scale with the modelled tariff and are illustrative; the shape of the trade-off does not.

14. Limitations

- **Sparse demand data.** Only five snapshot days exist, not a continuous record, so all annual figures are projections (mean daily saving $\times 365$), clearly labelled as such.
- **No job-level metadata.** The ORNL data records power, not job type or deadline, so shiftability is applied as a literature-backed 30% fraction rather than classified per job.
- **Single grid.** Results are specific to the nuclear-heavy TVA grid; a grid with higher renewable variability would offer deeper, though differently-shaped, clean windows.
- **Forecast difficulty.** Day-ahead carbon-intensity shape is hard to predict on TVA, limiting realised savings to roughly half the capacity-aware ceiling.
- **Static capacity model.** The load-shifting LP uses a representative per-node power and a fixed cluster capacity; a production system would use live node availability.
- **LSTM deferred.** The deep-learning model was not trained to completion on available hardware; its omission is documented rather than hidden.

15. Future Work

- **Shape-aware forecasting.** Develop and evaluate forecasters explicitly on the ordering of clean hours — for example seasonal models, direct multi-horizon learners, or hybrid level-plus-shape models — rather than on aggregate error.
- **Spatial shifting.** Extend temporal shifting with cross-regional migration by substituting TVA data with other US grids (for example MISO or CAISO) to estimate the additional saving from running workloads where the grid is cleanest, not only when.
- **Probabilistic scheduling.** Use forecast confidence bands to hedge scheduling decisions, deferring less aggressively when the forecast is uncertain — directly addressing the days on which the clean window flips.
- **Stochastic and robust optimisation.** The current schedulers are Linear and Mixed-Integer Programmes solved on a single point forecast. The strongest optimisation extension is stochastic programming, which would optimise over a distribution of possible carbon-intensity scenarios rather than betting on one forecast, directly hedging the days when the clean window flips; robust optimisation offers a lighter worst-case alternative that needs no scenario probabilities. Both require far more than five days of data. Duality analysis of the existing LP is a quick add-on whose shadow prices would reveal which clean hours are the binding capacity bottlenecks. Non-linear programming is not warranted at this scale, and reinforcement learning would be disproportionate for a feasibility study.
- **Live integration.** Connect the scheduler to a real batch system (SLURM/PBS) and live node-availability and grid feeds for a production pilot. This is also what would replace the 30% flexibility assumption with real job metadata (deadlines and durations a live scheduler already holds), turning the feasibility study into a deployable system.
- **Continuous demand data.** Acquire a continuous facility power trace to replace the five-day projection with a measured annual figure.

16. Conclusion

This project set out to test whether carbon-aware scheduling could reduce the Scope 2 emissions of a computing facility using real data, and to deliver the result as a usable tool. It succeeded on both counts, and did so with an unusual degree of honesty about what is and is not achievable.

The complete analytics lifecycle was built and validated: two authoritative data sources acquired and quality-assured; a carefully aligned hourly integrated dataset with audit-grade carbon intensity; a descriptive analysis that uncovered the pivotal moving-clean-window phenomenon; a rigorous forecasting comparison that selected XGBoost on point accuracy; a prescriptive optimisation that built greedy, LP and load-shifting schedulers; and a publicly deployed dashboard serving two stakeholders.

Three findings give the work its character. The clean window on the TVA grid moves unpredictably, so forecasting is essential. The best point-forecast model is not the best scheduling model, because scheduling depends on the shape of the day rather than the accuracy of the level. And real constraints — node capacity and forecast error — reduce the achievable saving from an optimistic 314 to a realistic 34 tonnes of CO₂ per year, a figure presented openly as part of a hierarchy rather than disguised.

The contribution is not a new algorithm but a rigorous, reproducible, honest pipeline from raw sensor data to a deployed decision-support tool — including a transferable lesson, rarely made explicit, that those building carbon-aware schedulers should select their forecasts on the ordering of clean hours, not on error alone. For an organisation facing CSRD disclosure, the result is a zero-capital, no-performance-penalty pathway to measurable and auditable carbon reduction, ready to replicate wherever delay-tolerant workloads run.

17. References

- Anasuri, S. and Pappula, K.K. (2023) 'Green HPC: Carbon-Aware Scheduling in Cloud Data Centers', IJERET, 4(2), pp. 106–114.
- Benhari, A. and Trystram, D. (2025) 'Adaptive Carbon-Aware Scheduling Policies for HPC Systems', Lecture Notes in Computer Science, 16210, pp. 1–15, Springer.
- EkkoSense (2025) 'ESG Data Centers — Climate Accountability Reporting for U.S. Data Centers'.
- Emergent Mind (2025) 'Carbon-Aware Scheduling — Survey of Frameworks'.
- European Commission (2024) 'Corporate Sustainability Reporting Directive (CSRD)'.
- IEA (2025) 'Energy and AI', International Energy Agency.
- Li, K. et al. (2025) 'More for Less: Integrating Capability-Predominant and Capacity-Predominant Computing', arXiv:2501.12464.
- Li, Z. et al. (2019) 'Thermal-Aware Hybrid Workload Management in a Green Datacenter Towards Renewable Energy Utilization', Energies, 12(8), pp. 1–18.
- Radovanovic, A. et al. (2023) 'Carbon-Aware Computing for Datacenters', IEEE Transactions on Power Systems, 38(2), pp. 1270–1280.
- Rodrigo, G.P. et al. (2018) 'Towards Understanding HPC Users and Systems: A NERSC Case Study', Journal of Parallel and Distributed Computing, 111, pp. 206–221.
- Shin, W. et al. (2021) 'Revealing Power, Energy and Thermal Dynamics of a 200PF Pre-Exascale Supercomputer', SC '21, ACM, Article 12, pp. 1–14.
- Vergara Larrea, V.G. et al. (2024) 'An HPC Co-Scheduler with Reinforcement Learning', arXiv:2401.09706.
- Verma, A. et al. (2015) 'Large-Scale Cluster Management at Google with Borg', EuroSys '15, ACM.

18. Appendix A — Project Structure and Reproducibility

The repository is organised into a configuration module, a data directory (raw and processed), a source package (acquisition, data, models, utilities), a pipelines runner, executed notebooks, diagrams, documentation and the deployed application.

HPC Optimizer/	
config/settings.py	single source of truth for constants & paths
data/raw/	5 Parquet + supply CSV (gitignored)
data/processed/	pipeline outputs + charts
src/acquisition/	EIA API fetcher
src/data/	demand, supply, integrator processors
src/models/	features, evaluation, forecaster, optimizer
pipelines/run_integration.py	one-command end-to-end runner
notebooks/01..05	executed analysis notebooks
app/main.py	Streamlit dashboard (two views)
docs/	PROJECT_GUIDE, DEPLOY, this report
requirements.txt	slim deployment deps
requirements-dev.txt	full pipeline deps

19. Appendix B — Decisions Log

Stage	Decision	Chosen	Rejected alternative
Acquisition	Demand source	ORNL Summit	Synthetic / cloud / NERSC
Acquisition	Supply source	EIA API v2	Electricity Maps / WattTime
Acquisition	Emission factors	IPCC AR5	EPA eGRID / commercial
Integration	Timezone	UTC→Eastern	Raw UTC join
Integration	Active classifier	GPU > 250 W	PSU power / node_state
Integration	Granularity	Hourly	Sub-hourly interpolation
Integration	Anomalies	Clip to 0	Drop rows
Integration	Missing values	Group-wise fill	Global fill / drop
Forecasting	Split	Chronological	Random split
Forecasting	Model	XGBoost	SARIMA / Prophet / LSTM
Optimisation	Baseline	Carbon-blind	Everything-at-hour-0
Optimisation	Fleet method	Load-shift LP	Unconstrained shift-to-min
Delivery	Platform	Streamlit	Power BI / Tableau

20. Appendix C — Key Configuration and Results

Parameter / Result	Value
Total nodes	4,626
GPU active threshold	250 W
Flexible fraction (sensitivity)	30% (20–40%)
Power per active node	≈ 1.2 kW
Mean grid carbon intensity	283 gCO ₂ /kWh (76–500)
Low-carbon grid share	56.3%
Mean cluster utilisation	76.5%
Selected forecast model	XGBoost (MAE 17.95)
Unconstrained annual saving	314 tCO ₂ /yr
Capacity-aware ceiling	87 tCO ₂ /yr
Realistic forecast-driven saving	34 tCO ₂ /yr
Hardware cost	€0

21. Appendix D — Data Dictionaries

This appendix documents the columns of the three processed datasets so that any reader can interpret the outputs without reading the source code.

D.1 demand_engineered.csv (120 rows)

Column	Type	Description
local_hour	datetime	Hour bucket in Eastern Prevailing Time
snapshot_date	date	One of the five ORNL observation days
hour_of_day	int	0–23 hour within the day
total_nodes	int	Nodes reporting that hour
active_nodes	int	Nodes with GPU power > 250 W (running a job)
idle_nodes	int	Nodes powered on but not running a job
utilization_rate	float	active_nodes / total_nodes
avg_gpu_power_W	float	Mean per-node total GPU power
cluster_power_kW	float	Total cluster power that hour
active_power_kW	float	Power attributable to active nodes
idle_power_kW	float	Standby power of idle nodes
active_energy_kWh	float	Active energy that hour (= active_power_kW × 1 h)
shiftable_energy_kWh	float	30% of active energy (central assumption)
shiftable_energy_20p_kWh	float	20% sensitivity
shiftable_energy_40p_kWh	float	40% sensitivity

D.2 supply_engineered.csv (35,064 rows) — selected columns

Column	Type	Description
datetime	datetime	Hour in Eastern Prevailing Time
COL, NG, NUC, OIL, OTH, SUN, WAT, WND	float	Generation by fuel type (MWh)
total_generation_MWh	float	Sum across all fuels
carbon_intensity_gCO2_per_kWh	float	Generation-weighted IPCC AR5 intensity
low_carbon_share	float	Clean generation / total
fossil_share	float	Fossil generation / total
hour_of_day, day_of_week, month	int	Calendar features
is_weekend	int	Weekend flag
season	str	Winter / Spring / Summer / Autumn
ci_rolling_24h, ci_rolling_7d	float	Rolling mean carbon intensity

Column	Type	Description
day_min_ci, day_max_ci	float	Daily minimum / maximum intensity

D.3 integrated.csv (120 rows) — selected columns

Column	Type	Description
carbon_intensity_gCO2_per_kWh	float	Grid intensity for that ORNL hour
day_min_ci	float	Cleanest hour's intensity that day
baseline_carbon_gCO2 / _kg	float	active_energy × actual intensity
optimized_carbon_gCO2 / _kg	float	After shifting 30% to day minimum
carbon_saved_gCO2 / _kg	float	Baseline – optimised (≥ 0)
carbon_reduction_pct	float	Saving as % of baseline
optimized_carbon_20p / 40p_gCO2	float	Sensitivity variants
is_green_hour	int	1 if intensity $\leq$ daily median
projected_annual_saving_tCO2	float	Mean daily saving × 365 (projection)

22. Appendix E — Worked Examples

E.1 Computing carbon intensity for one hour

Consider a single hour on the TVA grid with the following generation (MWh): Nuclear 8,204; Hydro 4,751; Natural Gas 1,684; Coal 1,672; Other 11; with Solar, Wind and Petroleum at zero. The total is 16,322 MWh. Applying the IPCC AR5 factors:

```
weighted = 1672*1000 + 1684*450 + 8204*0 + 4751*0 + 11*500
          = 1,672,000 + 757,800 + 0 + 0 + 5,500
          = 2,435,300 (gCO2 per MWh-equivalent units)
CI = 2,435,300 / 16,322 = 149.2 gCO2/kWh
```

This hour is unusually clean (149 versus the 283 mean) because nuclear and hydro together supply over 79% of generation and the only fossil contributions are modest coal and gas. The low-carbon share for this hour is $(8,204 + 4,751) / 16,322 = 79.4\%$.

E.2 A greedy scheduling decision

Suppose an operator submits a flexible job requiring 2,000 nodes for 4 hours, due within 24 hours, and the 48-hour forecast has its lowest 4-hour mean intensity at hours 14–18 (300.9 gCO₂/kWh) versus 311.0 for running immediately at hours 0–4. The job's energy is:

```
energy = nodes x power_per_node x duration
        = 2000 x 1.2 kW x 4 h = 9,600 kWh
```

Running now would emit $9,600 \times 311.0 / 1000 = 2,985.6$ kg CO₂; deferring to hours 14–18 emits $9,600 \times 300.9 / 1000 = 2,888.6$ kg. The saving is 97.0 kg CO₂ for identical computation — about 3.3% — achieved purely by waiting fourteen hours. This is exactly what the dashboard's scheduler widget returns.

E.3 Why capacity limits the fleet saving

On 20 January 2020 the cluster's active energy peaked near 5,290 kWh in an hour, against a capacity ceiling of $4,626 \text{ nodes} \times 1.2 \text{ kW} = 5,551 \text{ kWh}$. With roughly 70% of each hour's load fixed (non-shiftable), a clean evening hour carrying 2,800 kWh of fixed load can absorb only about 2,750 kWh of incoming shifted energy before hitting the ceiling. Across the few clean hours within reach, only about 11,091 kWh of the day's 30,477 kWh of shiftable energy could actually be moved — which is why the realistic saving is far below the unconstrained figure that assumed unlimited absorption.

23. Appendix F — Glossary

Active node. A node whose total GPU power exceeds 250 W, taken to indicate a running job.

Baseboard Management Controller (BMC). Embedded controller in each node that measures PSU input power independently of the operating system.

Balancing authority. An entity responsible for matching electricity supply and demand on a grid; TVA is one.

Carbon intensity (CI). Grams of CO₂ emitted per kilowatt-hour of electricity; varies hourly with the generation mix.

Carbon-aware scheduling. Scheduling workloads with awareness of grid carbon intensity to run them when electricity is cleanest.

CSRD. EU Corporate Sustainability Reporting Directive, mandating Scope 1–3 emissions disclosure from 2025.

DCGM. NVIDIA Data Center GPU Manager, which reads per-GPU power from on-board sensors.

Delay-tolerant workload. A job that can be deferred without harm, e.g. a batch simulation, backup or training run.

Greedy heuristic. A scheduler that places each job in the lowest-carbon feasible window one job at a time.

Linear Programme (LP) / MIP. A mathematical optimisation that minimises an objective subject to linear constraints; MIP adds integer/binary variables.

Rolling-origin evaluation. Forecasting from many successive origins and pooling errors, used to assess multi-step forecast accuracy honestly.

Scope 2 emissions. Indirect emissions from purchased electricity — the dominant footprint for data centres.

Shiftable energy. The portion of active energy assumed flexible enough to move to a cleaner hour (30% centrally).

Temporal shifting. Moving a workload to a different time to reduce its carbon, the core mechanism of this project.